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(54) **LABELING DETECTED OBJECTS IN
FRAMES OF A VIDEO**

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H04N 19/70 (2014.01)
G06V 20/40 (2022.01)

(Continued)

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CPC **H04N 19/70** (2014.11); **G06V 20/41**
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19/172 (2014.11)

(58) **Field of Classification Search**
CPC H04N 19/70; H04N 19/167; H04N 19/172;
H04N 21/2353; G06V 20/41
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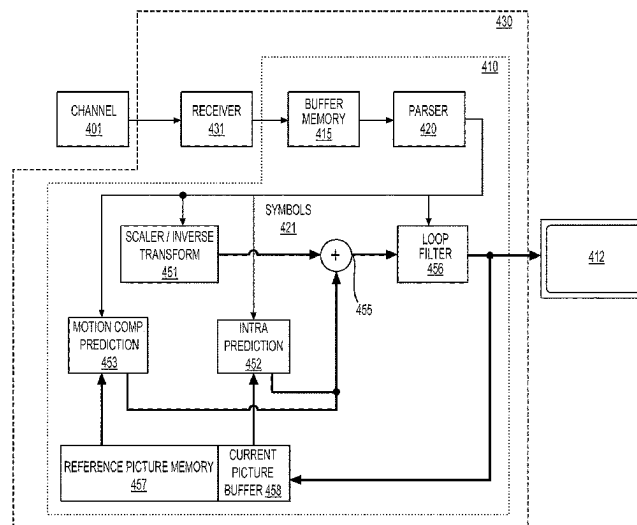
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(57) **ABSTRACT**

Aspects of the disclosure include methods, apparatuses, and
non-transitory computer-readable storage mediums for
video encoding/decoding. An apparatus includes processing
circuitry that receives metadata associated with a coded
video bitstream. The metadata includes labeling information
of one or more objects detected in a first picture that is coded
in the coded video bitstream. The processing circuitry
decodes the labeling information of the one or more objects
in the first picture that is coded in the coded video bitstream.
The processing circuitry applies the labeling information to
the one or more objects in the first picture.

17 Claims, 13 Drawing Sheets



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(51) **Int. Cl.**

H04N 19/167 (2014.01)

H04N 19/172 (2014.01)

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	R01	R02	R03	R04	R05	R06	R07	R08	R09
R10	<u>S11</u>	S12	S13	S14	104				
<u>R20</u>	S21	S22	S23	<u>S24</u>					
R30	S31	<u>S32</u>	S33	S34					
R40	<u>S41</u>	S42	S43	<u>S44</u>					
R50									
<u>R60</u>									
R70									

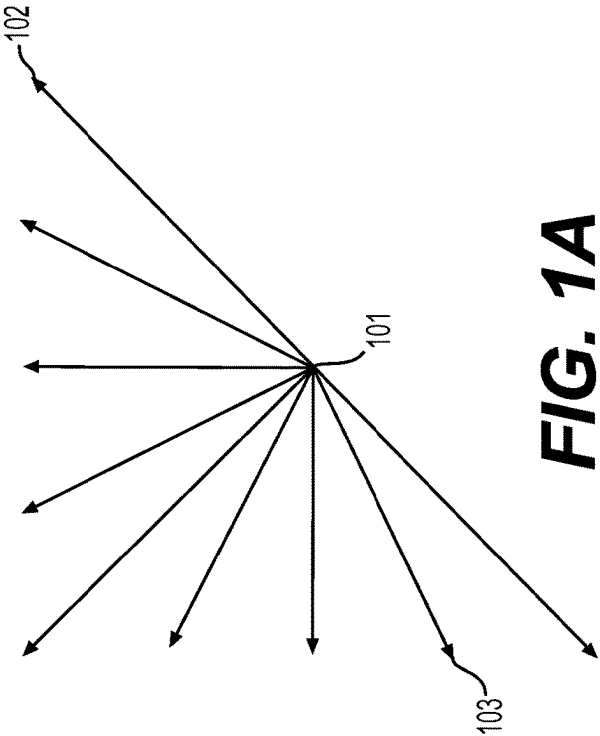
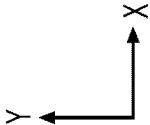
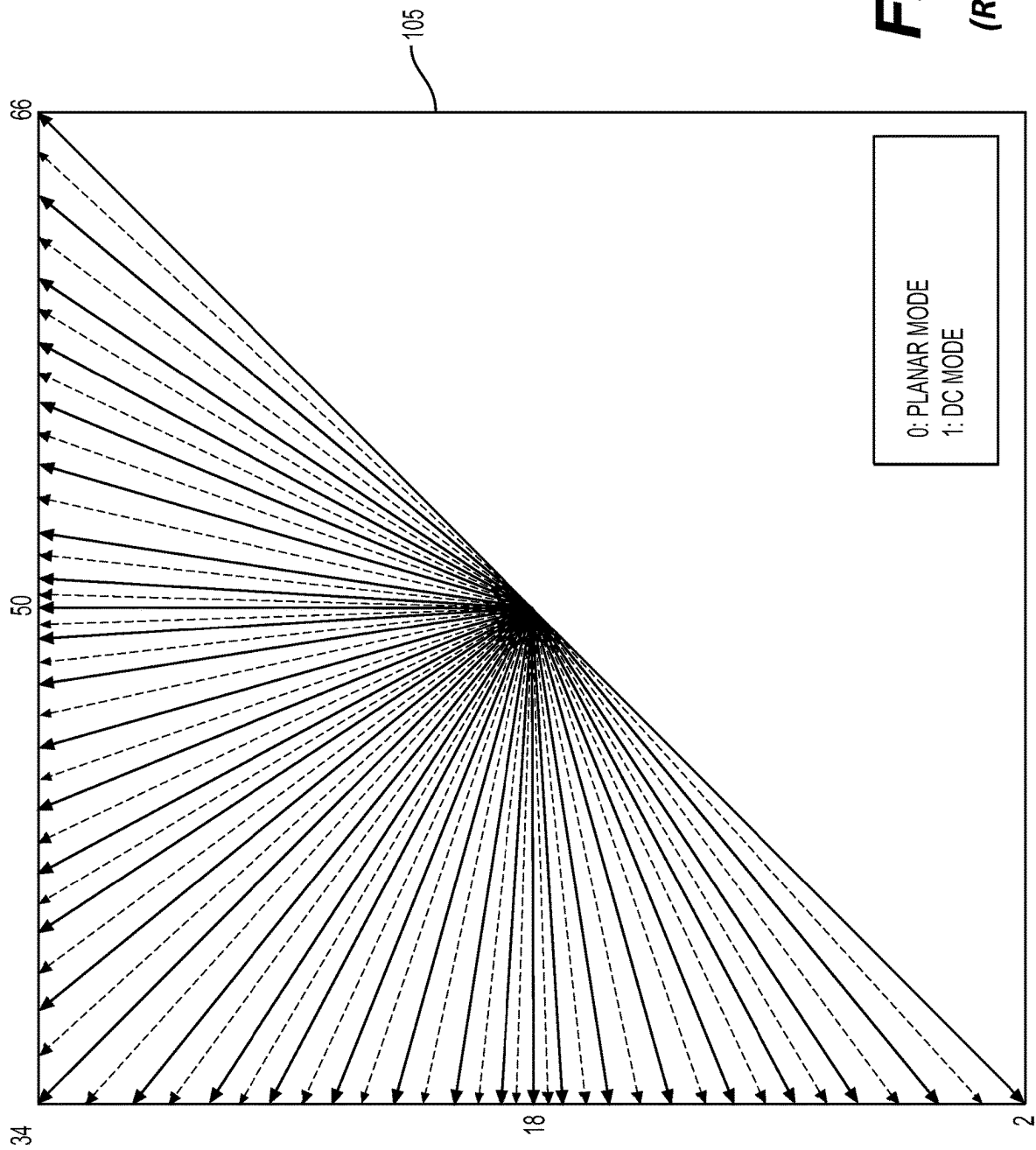


FIG. 1A
(Related Art)



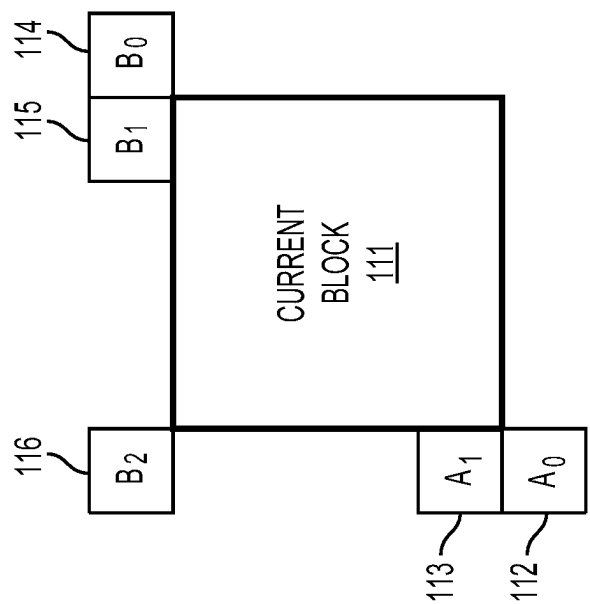


FIG. 1C
(Related Art)

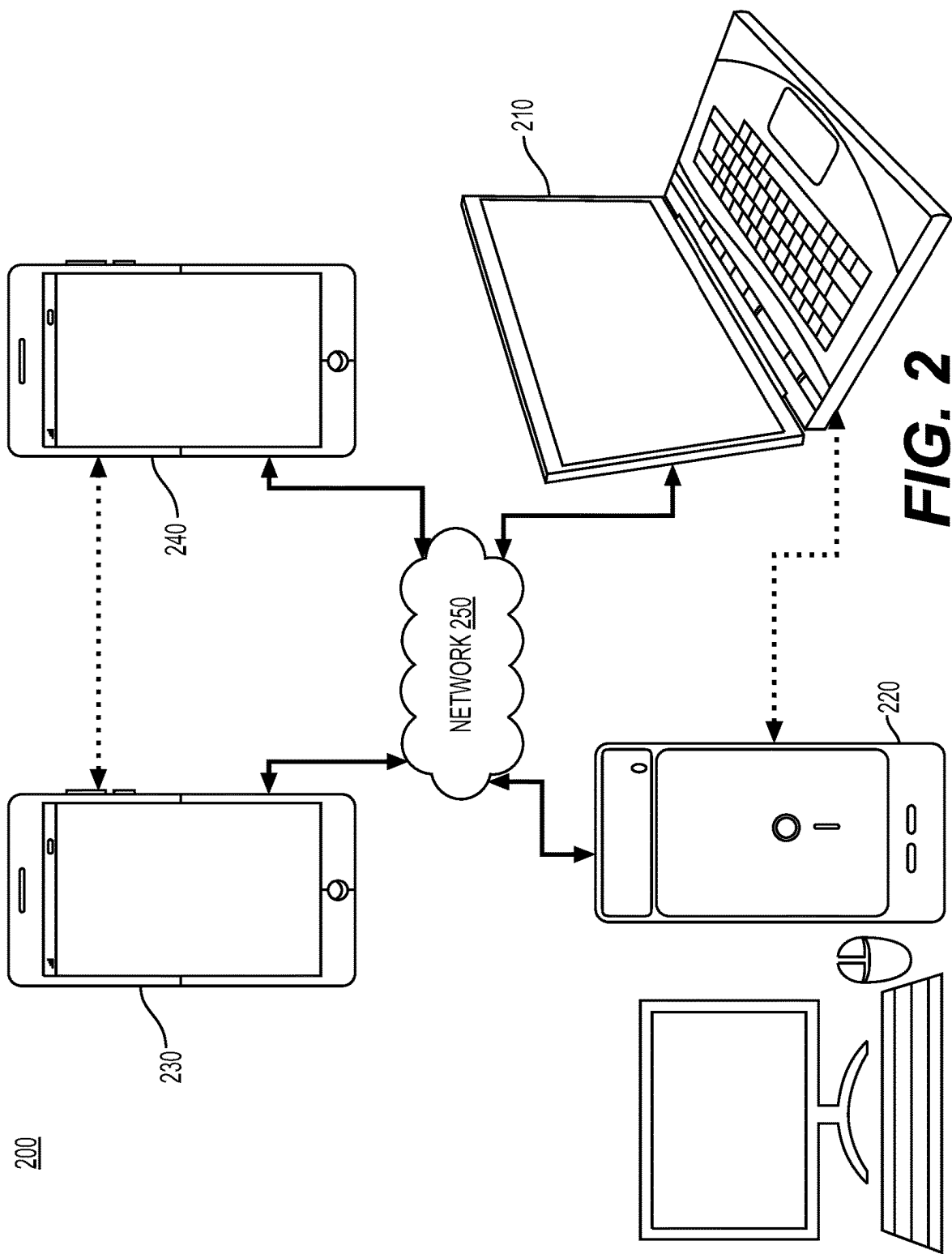


FIG. 2

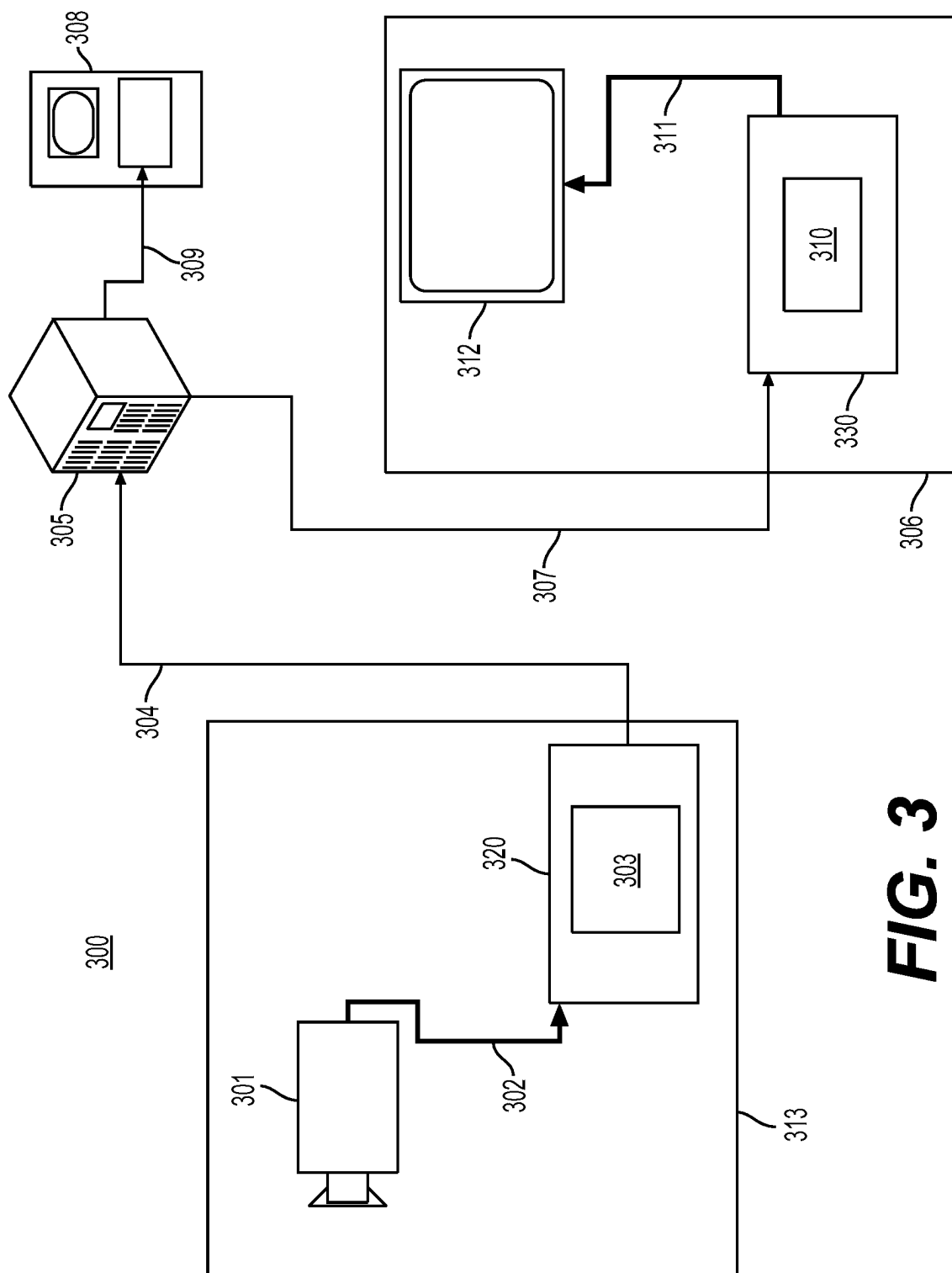


FIG. 3

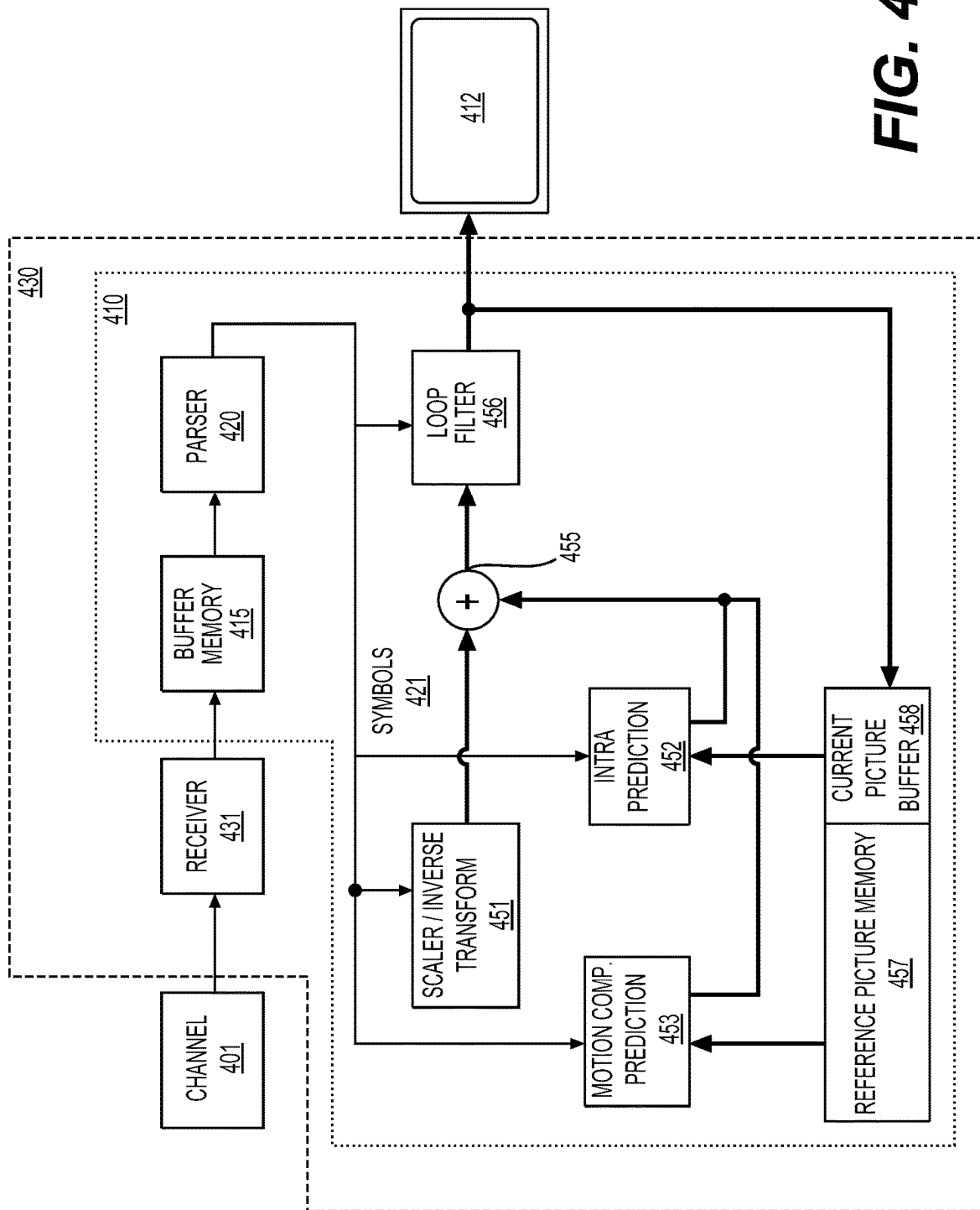
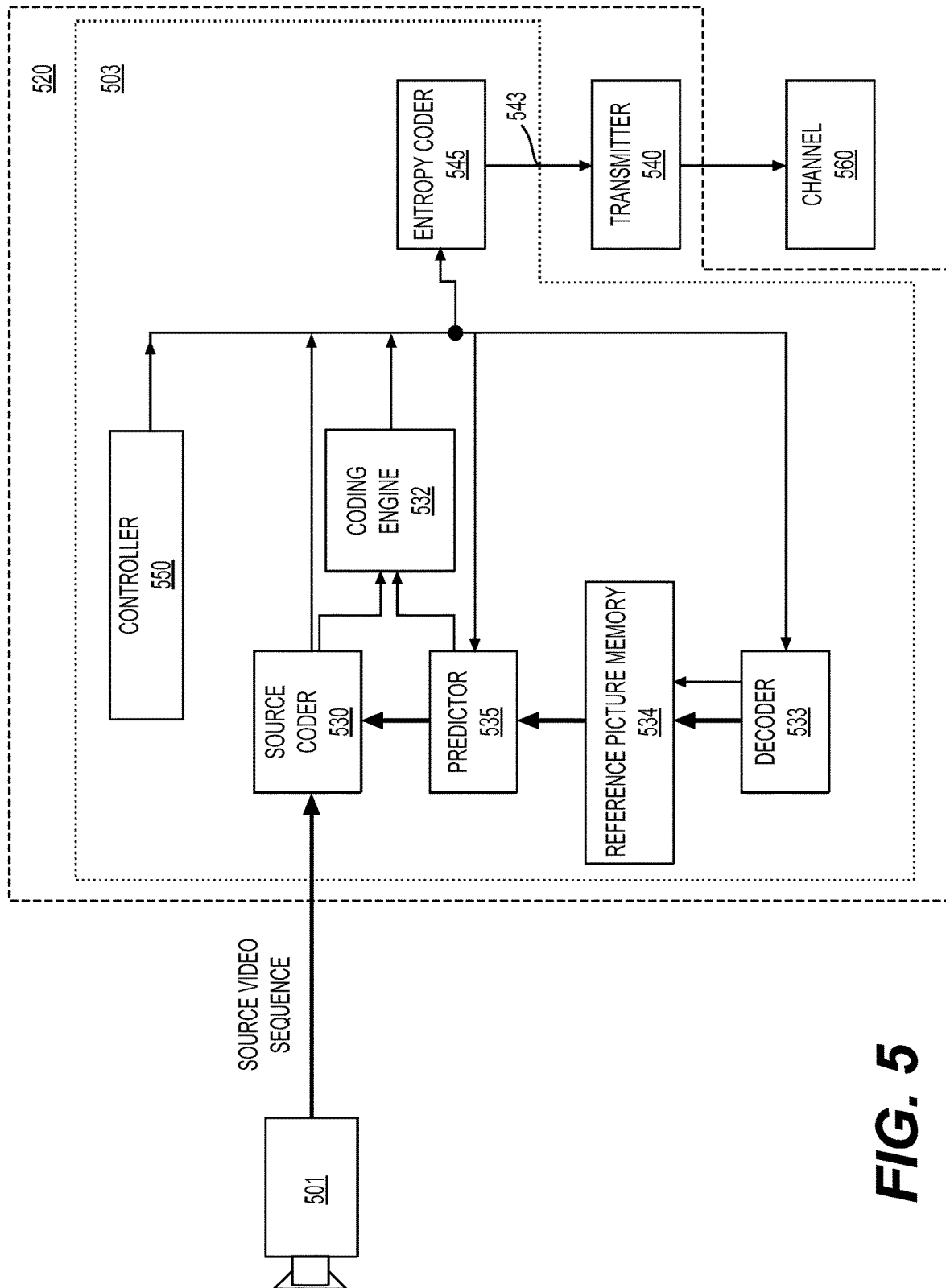
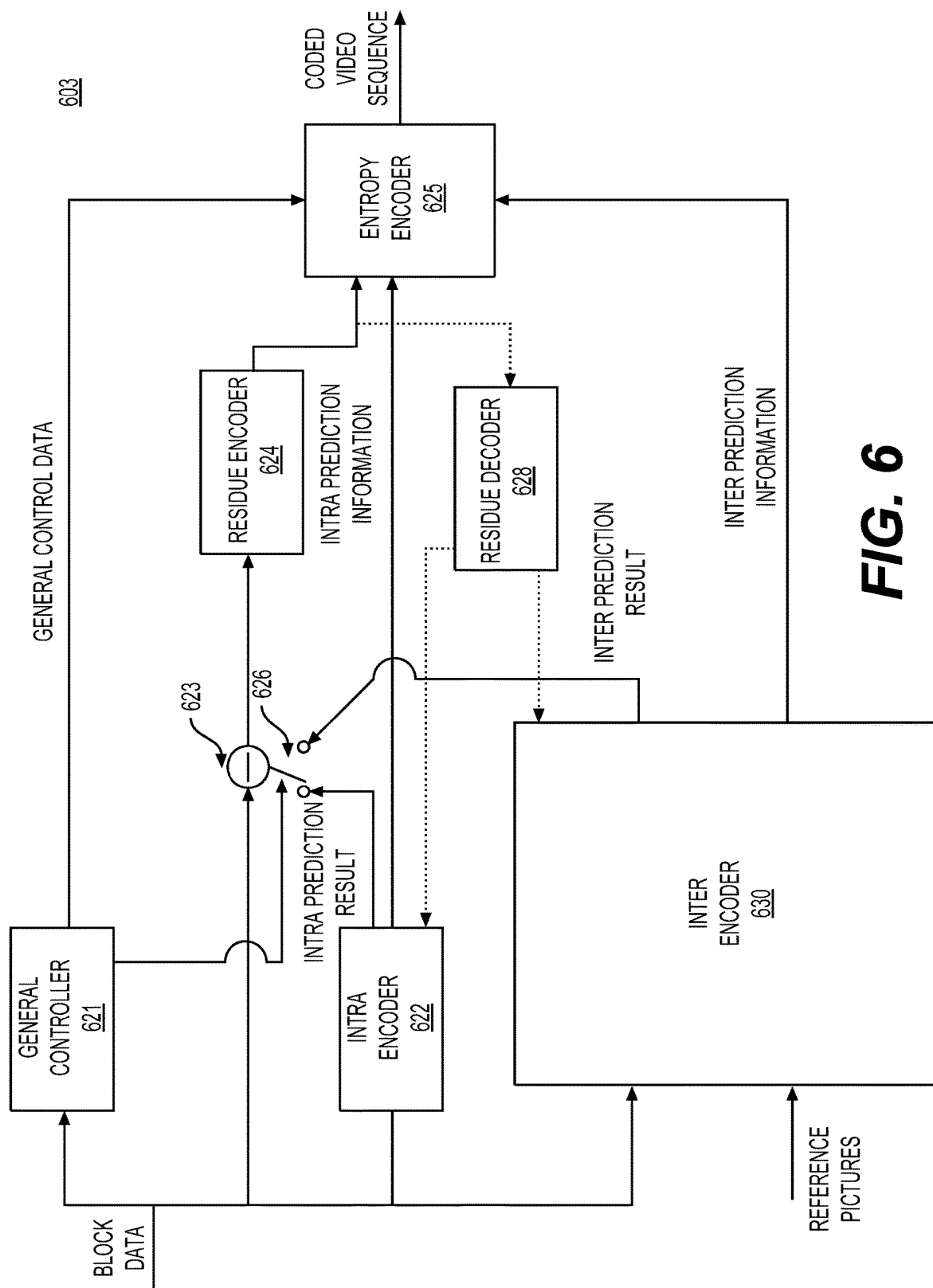


FIG. 4





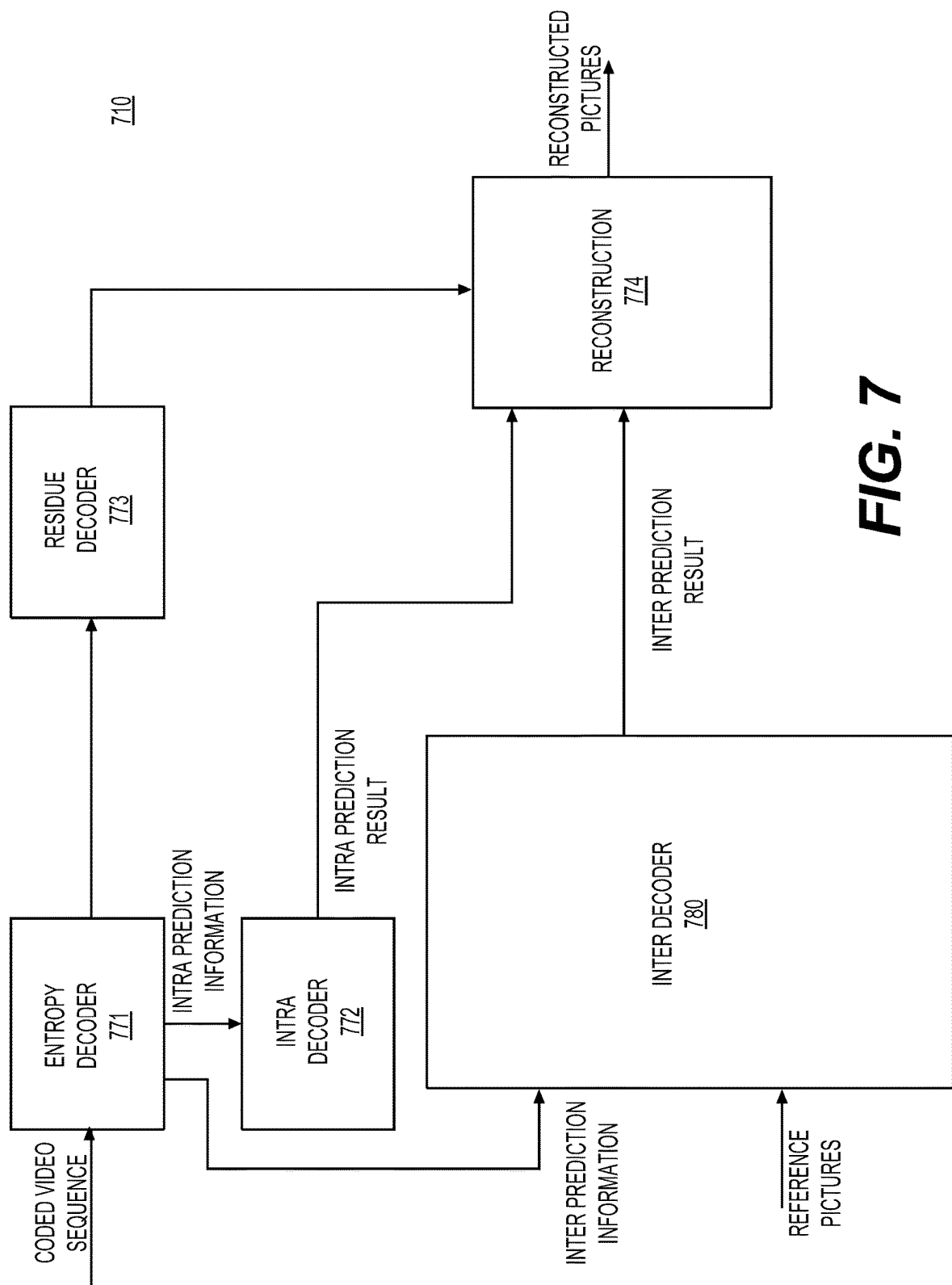


FIG. 7

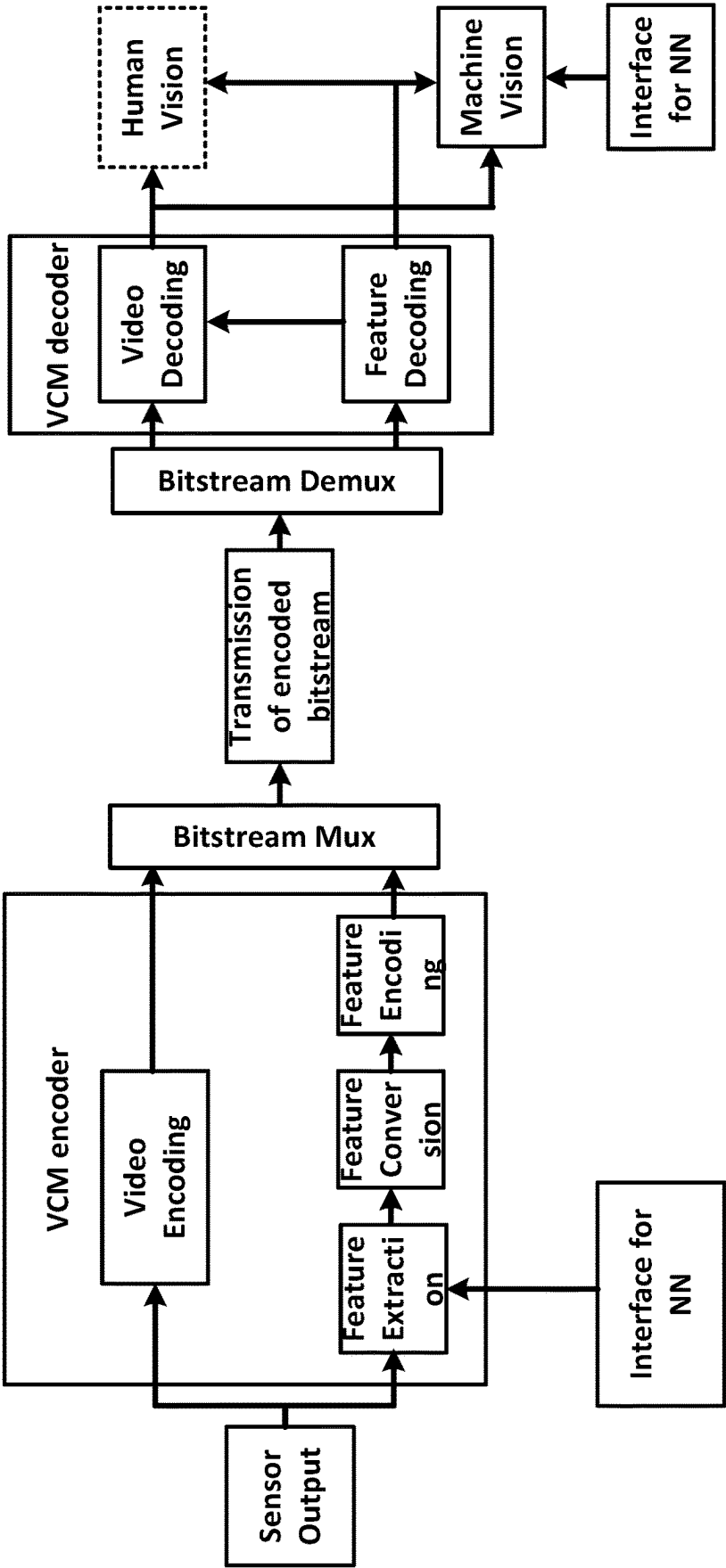
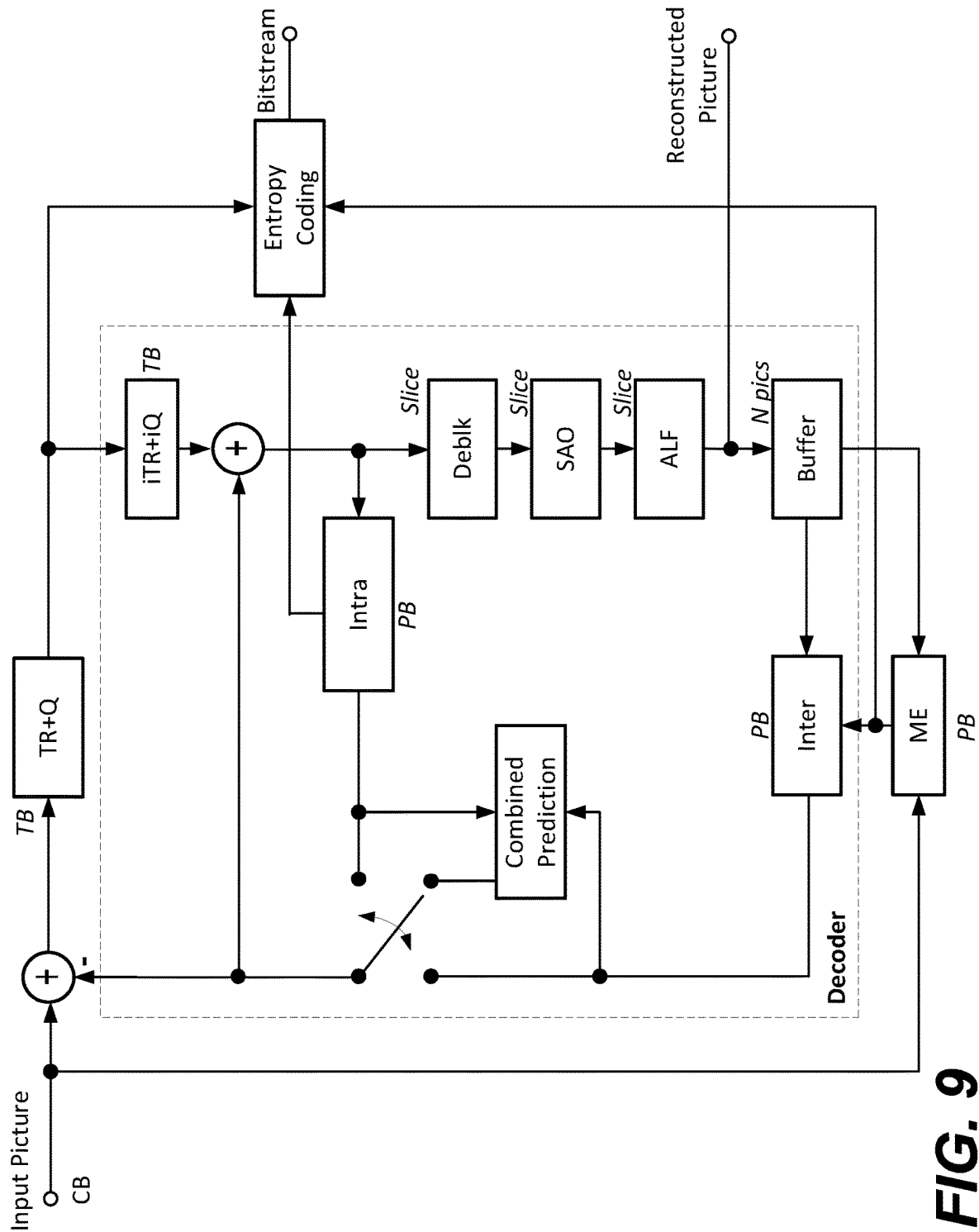
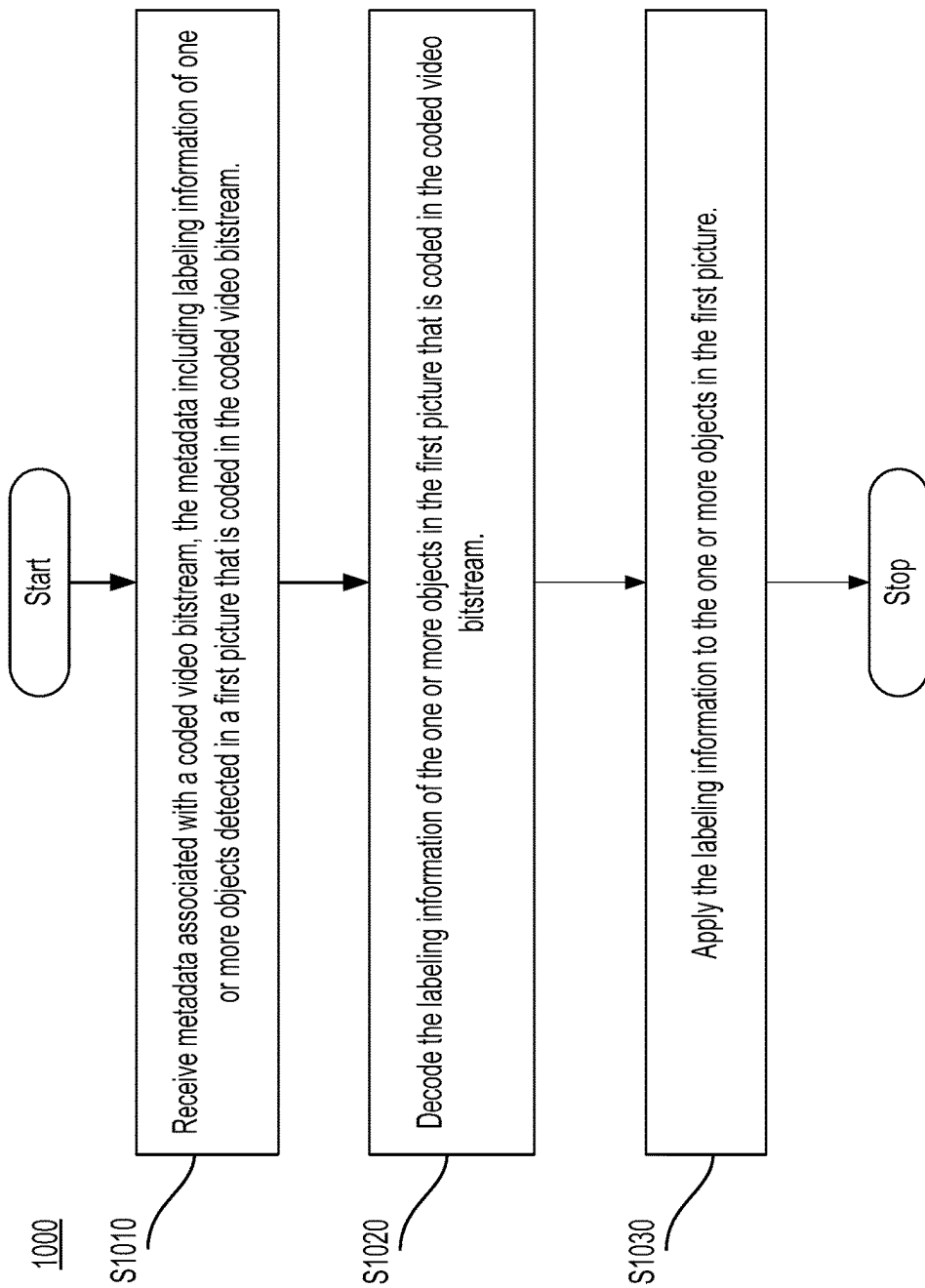


FIG. 8



**FIG. 10**

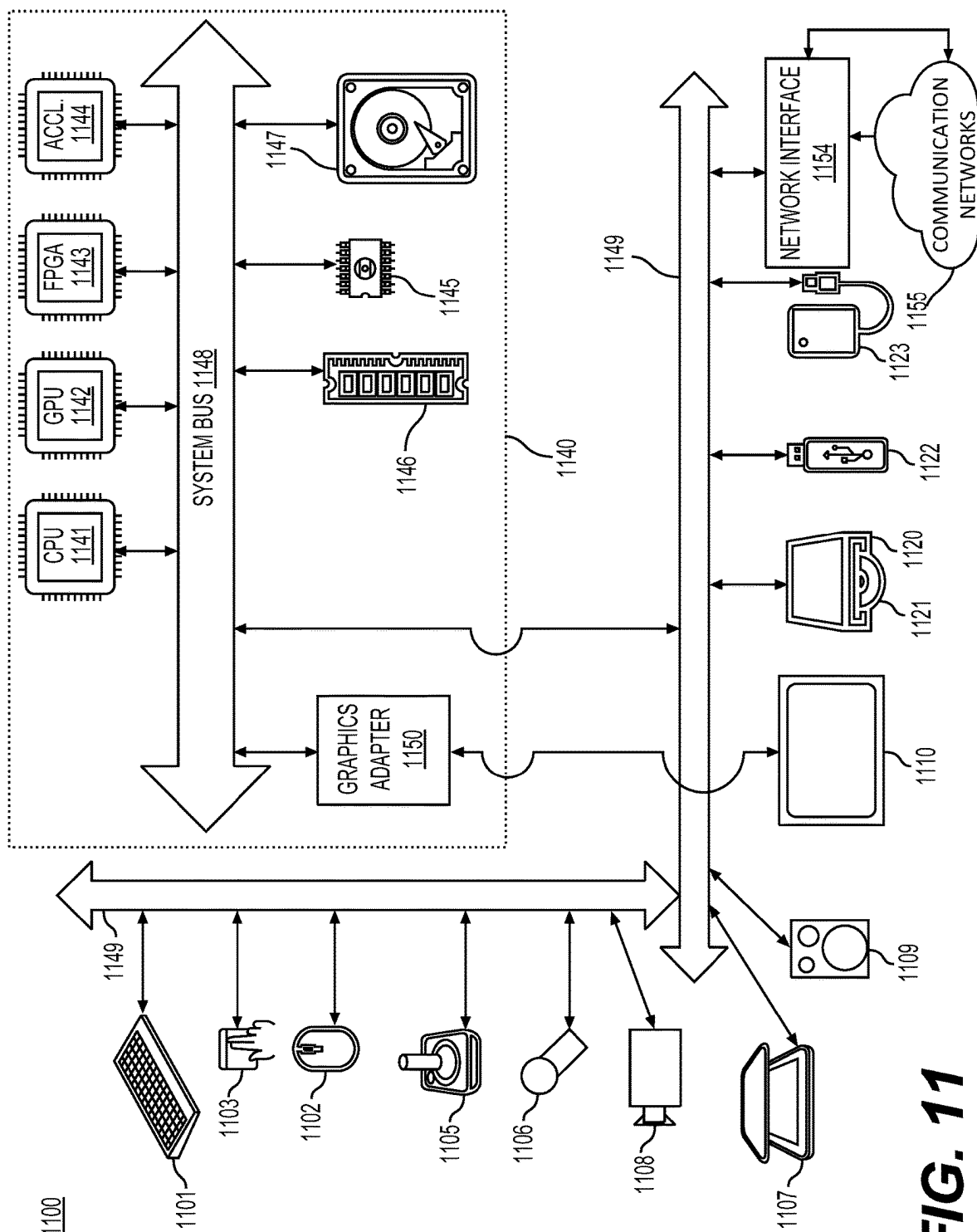


FIG. 11

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LABELING DETECTED OBJECTS IN FRAMES OF A VIDEO

INCORPORATION BY REFERENCE

This present application is a continuation of U.S. application Ser. No. 17/459,753, "METHOD AND APPARATUS FOR VIDEO CODING" filed on Aug. 27, 2021, which claims the benefit of priority to U.S. Provisional Application No. 63/135,530, "SIGNALING OF OBJECTS FOR MACHINE TASKS," filed on Jan. 8, 2021, which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

The present disclosure describes embodiments generally related to video coding.

BACKGROUND

The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent the work is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

Video coding and decoding can be performed using inter-picture prediction with motion compensation. Uncompressed digital video can include a series of pictures, each picture having a spatial dimension of, for example, 1920×1080 luminance samples and associated chrominance samples. The series of pictures can have a fixed or variable picture rate (informally also known as frame rate) of, for example, 60 pictures per second or 60 Hz. Uncompressed video has significant bitrate requirements. For example, 1080p60 4:2:0 video at 8 bit per sample (1920×1080 luminance sample resolution at 60 Hz frame rate) requires close to 1.5 Gbit/s bandwidth. An hour of such video requires more than 600 GBytes of storage space.

One purpose of video coding and decoding can be the reduction of redundancy in the input video signal, through compression. Compression can help reduce the aforementioned bandwidth or storage space requirements, in some cases by two orders of magnitude or more. Both lossless and lossy compression, as well as a combination thereof can be employed. Lossless compression refers to techniques where an exact copy of the original signal can be reconstructed from the compressed original signal. When using lossy compression, the reconstructed signal may not be identical to the original signal, but the distortion between original and reconstructed signals is small enough to make the reconstructed signal useful for the intended application. In the case of video, lossy compression is widely employed. The amount of distortion tolerated depends on the application; for example, users of certain consumer streaming applications may tolerate higher distortion than users of television distribution applications. The compression ratio achievable can reflect that: higher allowable/tolerable distortion can yield higher compression ratios.

A video encoder and decoder can utilize techniques from several broad categories, including, for example, motion compensation, transform, quantization, and entropy coding.

Video codec technologies can include techniques known as intra coding. In intra coding, sample values are represented without reference to samples or other data from

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previously reconstructed reference pictures. In some video codecs, the picture is spatially subdivided into blocks of samples. When all blocks of samples are coded in intra mode, that picture can be an intra picture. Intra pictures and their derivations such as independent decoder refresh pictures, can be used to reset the decoder state and can, therefore, be used as the first picture in a coded video bitstream and a video session, or as a still image. The samples of an intra block can be exposed to a transform, and the transform coefficients can be quantized before entropy coding. Intra prediction can be a technique that minimizes sample values in the pre-transform domain. In some cases, the smaller the DC value after a transform is, and the smaller the AC coefficients are, the fewer the bits that are required at a given quantization step size to represent the block after entropy coding.

Traditional intra coding such as known from, for example MPEG-2 generation coding technologies, does not use intra prediction. However, some newer video compression technologies include techniques that attempt, from, for example, surrounding sample data and/or metadata obtained during the encoding and/or decoding of spatially neighboring, and preceding in decoding order, blocks of data. Such techniques are henceforth called "intra prediction" techniques. Note that in at least some cases, intra prediction is only using reference data from the current picture under reconstruction and not from reference pictures.

There can be many different forms of intra prediction. When more than one of such techniques can be used in a given video coding technology, the technique in use can be coded in an intra prediction mode. In certain cases, modes can have submodes and/or parameters, and those can be coded individually or included in the mode codeword. Which codeword to use for a given mode, submode, and/or parameter combination can have an impact in the coding efficiency gain through intra prediction, and so can the entropy coding technology used to translate the codewords into a bitstream.

A certain mode of intra prediction was introduced with H.264, refined in H.265, and further refined in newer coding technologies such as joint exploration model (JEM), versatile video coding (VVC), and benchmark set (BMS). A predictor block can be formed using neighboring sample values belonging to already available samples. Sample values of neighboring samples are copied into the predictor block according to a direction. A reference to the direction in use can be coded in the bitstream or may be predicted itself.

Referring to FIG. 1A, depicted in the lower right is a subset of nine predictor directions known from H.265's 33 possible predictor directions (corresponding to the 33 angular modes of the 35 intra modes). The point where the arrows converge (101) represents the sample being predicted. The arrows represent the direction from which the sample is being predicted. For example, arrow (102) indicates that sample (101) is predicted from a sample or samples to the upper right, at a 45 degree angle from the horizontal. Similarly, arrow (103) indicates that sample (101) is predicted from a sample or samples to the lower left of sample (101), in a 22.5 degree angle from the horizontal.

Still referring to FIG. 1A, on the top left there is depicted a square block (104) of 4×4 samples (indicated by a dashed, boldface line). The square block (104) includes 16 samples, each labelled with an "S", its position in the Y dimension (e.g., row index) and its position in the X dimension (e.g., column index). For example, sample S21 is the second sample in the Y dimension (from the top) and the first (from

the left) sample in the X dimension. Similarly, sample **S44** is the fourth sample in block **(104)** in both the Y and X dimensions. As the block is 4×4 samples in size, **S44** is at the bottom right. Further shown are reference samples that follow a similar numbering scheme. A reference sample is labelled with an R, its Y position (e.g., row index) and X position (column index) relative to block **(104)**. In both H.264 and H.265, prediction samples neighbor the block under reconstruction; therefore no negative values need to be used.

Intra picture prediction can work by copying reference sample values from the neighboring samples as appropriated by the signaled prediction direction. For example, assume the coded video bitstream includes signaling that, for this block, indicates a prediction direction consistent with arrow **(102)**—that is, samples are predicted from a prediction sample or samples to the upper right, at a 45 degree angle from the horizontal. In that case, samples **S41**, **S32**, **S23**, and **S14** are predicted from the same reference sample **R05**. Sample **S44** is then predicted from reference sample **R08**.

In certain cases, the values of multiple reference samples may be combined, for example through interpolation, in order to calculate a reference sample; especially when the directions are not evenly divisible by 45 degrees.

The number of possible directions has increased as video coding technology has developed. In H.264 (year 2003), nine different direction could be represented. That increased to 33 in H.265 (year 2013), and JEM/VVC/BMS, at the time of disclosure, can support up to 65 directions. Experiments have been conducted to identify the most likely directions, and certain techniques in the entropy coding are used to represent those likely directions in a small number of bits, accepting a certain penalty for less likely directions. Further, the directions themselves can sometimes be predicted from neighboring directions used in neighboring, already decoded, blocks.

FIG. 1B shows a schematic **(105)** that depicts 65 intra prediction directions according to JEM to illustrate the increasing number of prediction directions over time.

The mapping of intra prediction directions bits in the coded video bitstream that represent the direction can be different from video coding technology to video coding technology; and can range, for example, from simple direct mappings of prediction direction to intra prediction mode, to codewords, to complex adaptive schemes involving most probable modes, and similar techniques. In all cases, however, there can be certain directions that are statistically less likely to occur in video content than certain other directions. As the goal of video compression is the reduction of redundancy, those less likely directions will, in a well working video coding technology, be represented by a larger number of bits than more likely directions.

Motion compensation can be a lossy compression technique and can relate to techniques where a block of sample data from a previously reconstructed picture or part thereof (reference picture), after being spatially shifted in a direction indicated by a motion vector (MV henceforth), is used for the prediction of a newly reconstructed picture or picture part. In some cases, the reference picture can be the same as the picture currently under reconstruction. MVs can have two dimensions X and Y, or three dimensions, the third being an indication of the reference picture in use (the latter, indirectly, can be a time dimension).

In some video compression techniques, an MV applicable to a certain area of sample data can be predicted from other MVs, for example from those related to another area of sample data spatially adjacent to the area under reconstruction,

and preceding that MV in decoding order. Doing so can substantially reduce the amount of data required for coding the MV, thereby removing redundancy and increasing compression. MV prediction can work effectively, for example, because when coding an input video signal derived from a camera (known as natural video) there is a statistical likelihood that areas larger than the area to which a single MV is applicable move in a similar direction and, therefore, can in some cases be predicted using a similar MV derived from MVs of a neighboring area. That results in the MV found for a given area to be similar or the same as the MV predicted from the surrounding MVs, and that in turn can be represented, after entropy coding, in a smaller number of bits than what would be used if coding the MV directly. In some cases, MV prediction can be an example of lossless compression of a signal (namely: the MVs) derived from the original signal (namely: the sample stream). In other cases, MV prediction itself can be lossy, for example because of rounding errors when calculating a predictor from several surrounding MVs.

Various MV prediction mechanisms are described in H.265/HEVC (ITU-T Rec. H.265, “High Efficiency Video Coding”, December 2016). Out of the many MV prediction mechanisms that H.265 offers, described herein is a technique henceforth referred to as “spatial merge.”

Referring to FIG. 1C, a current block **(111)** can include samples that have been found by the encoder during the motion search process to be predictable from a previous block of the same size that has been spatially shifted. Instead of coding that MV directly, the MV can be derived from metadata associated with one or more reference pictures, for example from the most recent (in decoding order) reference picture, using the MV associated with either one of five surrounding samples, denoted A0, A1, and B0, B1, B2 **(112 through 116, respectively)**. In H.265, the MV prediction can use predictors from the same reference picture that the neighboring block is using.

SUMMARY

Aspects of the disclosure provide apparatuses for video encoding/decoding. An apparatus includes processing circuitry that receives metadata associated with a coded video bitstream. The metadata includes labeling information of one or more objects detected in a first picture that is coded in the coded video bitstream. The processing circuitry decodes the labeling information of the one or more objects in the first picture that is coded in the coded video bitstream. The processing circuitry applies the labeling information to the one or more objects in the first picture.

In one embodiment, the metadata is included in a supplementary enhancement information (SEI) message in the coded video bitstream.

In one embodiment, the metadata is included in a file that is separate from the coded video bitstream.

In one embodiment, the labeling information indicates a total number of bounding boxes in the first picture and includes location information and size information of each bounding box, each bounding box being associated with one of the one or more objects in the first picture.

In one embodiment, the labeling information includes category information that indicates a category for each of the one or more objects.

In one embodiment, the labeling information includes identification information that identifies each of the one or more objects in a video sequence.

In one embodiment, the location information of one of the bounding boxes includes a location offset of the one of the bounding boxes between the first picture and a second picture that is coded in the video bitstream.

In one embodiment, the location information of one of the bounding boxes indicates a location outside the first picture for the one of the bounding boxes based on an object associated with the one of the bounding boxes not existing in the first picture.

In one embodiment, the processing circuitry sends a request to receive the metadata associated with the coded video bitstream.

Aspects of the disclosure provide methods for video encoding/decoding. The methods can perform any one or a combination of the processes performed by the apparatuses for video encoding/decoding. In the method, metadata associated with a coded video bitstream is received. The metadata includes labeling information of one or more objects detected in a first picture that is coded in the coded video bitstream. The labeling information of the one or more objects in the first picture that is coded in the coded video bitstream is decoded. The labeling information is applied to the one or more objects in the first picture.

Aspects of the disclosure also provide non-transitory computer-readable mediums storing instructions which when executed by at least one processor cause the at least one processor to perform any one or a combination of the methods for video encoding/decoding.

BRIEF DESCRIPTION OF THE DRAWINGS

Further features, the nature, and various advantages of the disclosed subject matter will be more apparent from the following detailed description and the accompanying drawings in which:

FIG. 1A is a schematic illustration of an exemplary subset of intra prediction modes;

FIG. 1B is an illustration of exemplary intra prediction directions;

FIG. 1C is a schematic illustration of a current block and its surrounding spatial merge candidates in one example;

FIG. 2 is a schematic illustration of a simplified block diagram of a communication system in accordance with an embodiment;

FIG. 3 is a schematic illustration of a simplified block diagram of a communication system in accordance with an embodiment;

FIG. 4 is a schematic illustration of a simplified block diagram of a decoder in accordance with an embodiment;

FIG. 5 is a schematic illustration of a simplified block diagram of an encoder in accordance with an embodiment;

FIG. 6 shows a block diagram of an encoder in accordance with another embodiment;

FIG. 7 shows a block diagram of a decoder in accordance with another embodiment;

FIG. 8 shows an exemplary architecture used in video coding for machine (VCM) in accordance with an embodiment;

FIG. 9 shows an exemplary encoder used in versatile video coding (VVC) in accordance with an embodiment;

FIG. 10 shows an exemplary flowchart in accordance with an embodiment; and

FIG. 11 is a schematic illustration of a computer system in accordance with an embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

I. Video Decoder and Encoder Systems

FIG. 2 illustrates a simplified block diagram of a communication system (200) according to an embodiment of the present disclosure. The communication system (200) includes a plurality of terminal devices that can communicate with each other, via, for example, a network (250). For example, the communication system (200) includes a first pair of terminal devices (210) and (220) interconnected via the network (250). In the FIG. 2 example, the first pair of terminal devices (210) and (220) performs unidirectional transmission of data. For example, the terminal device (210) may code video data (e.g., a stream of video pictures that are captured by the terminal device (210)) for transmission to the other terminal device (220) via the network (250). The encoded video data can be transmitted in the form of one or more coded video bitstreams. The terminal device (220) may receive the coded video data from the network (250), decode the coded video data to recover the video pictures and display video pictures according to the recovered video data. Unidirectional data transmission may be common in media serving applications and the like.

In another example, the communication system (200) includes a second pair of terminal devices (230) and (240) that performs bidirectional transmission of coded video data that may occur, for example, during videoconferencing. For bidirectional transmission of data, in an example, each terminal device of the terminal devices (230) and (240) may code video data (e.g., a stream of video pictures that are captured by the terminal device) for transmission to the other terminal device of the terminal devices (230) and (240) via the network (250). Each terminal device of the terminal devices (230) and (240) also may receive the coded video data transmitted by the other terminal device of the terminal devices (230) and (240), and may decode the coded video data to recover the video pictures and may display video pictures at an accessible display device according to the recovered video data.

In the FIG. 2 example, the terminal devices (210), (220), (230) and (240) may be illustrated as servers, personal computers and smart phones but the principles of the present disclosure may be not so limited. Embodiments of the present disclosure find application with laptop computers, tablet computers, media players and/or dedicated video conferencing equipment. The network (250) represents any number of networks that convey coded video data among the terminal devices (210), (220), (230) and (240), including for example wireline (wired) and/or wireless communication networks. The communication network (250) may exchange data in circuit-switched and/or packet-switched channels. Representative networks include telecommunications networks, local area networks, wide area networks and/or the Internet. For the purposes of the present discussion, the architecture and topology of the network (250) may be immaterial to the operation of the present disclosure unless explained herein below.

FIG. 3 illustrates, as an example for an application for the disclosed subject matter, the placement of a video encoder and a video decoder in a streaming environment. The disclosed subject matter can be equally applicable to other video enabled applications, including, for example, video conferencing, digital TV, storing of compressed video on digital media including CD, DVD, memory stick, and the like.

A streaming system may include a capture subsystem (313) that can include a video source (301), for example a

digital camera, creating for example a stream of video pictures (302) that are uncompressed. In an example, the stream of video pictures (302) includes samples that are taken by the digital camera. The stream of video pictures (302), depicted as a bold line to emphasize a high data volume when compared to encoded video data (304) (or coded video bitstreams), can be processed by an electronic device (320) that includes a video encoder (303) coupled to the video source (301). The video encoder (303) can include hardware, software, or a combination thereof to enable or implement aspects of the disclosed subject matter as described in more detail below. The encoded video data (304) (or encoded video bitstream (304)), depicted as a thin line to emphasize the lower data volume when compared to the stream of video pictures (302), can be stored on a streaming server (305) for future use. One or more streaming client subsystems, such as client subsystems (306) and (308) in FIG. 3 can access the streaming server (305) to retrieve copies (307) and (309) of the encoded video data (304). A client subsystem (306) can include a video decoder (310), for example, in an electronic device (330). The video decoder (310) decodes the incoming copy (307) of the encoded video data and creates an outgoing stream of video pictures (311) that can be rendered on a display (312) (e.g., display screen) or other rendering device (not depicted). In some streaming systems, the encoded video data (304), (307), and (309) (e.g., video bitstreams) can be encoded according to certain video coding/compression standards. Examples of those standards include ITU-T Recommendation H.265. In an example, a video coding standard under development is informally known as Versatile Video Coding (VVC). The disclosed subject matter may be used in the context of VVC.

It is noted that the electronic devices (320) and (330) can include other components (not shown). For example, the electronic device (320) can include a video decoder (not shown) and the electronic device (330) can include a video encoder (not shown) as well.

FIG. 4 shows a block diagram of a video decoder (410) according to an embodiment of the present disclosure. The video decoder (410) can be included in an electronic device (430). The electronic device (430) can include a receiver (431) (e.g., receiving circuitry). The video decoder (410) can be used in the place of the video decoder (310) in the FIG. 3 example.

The receiver (431) may receive one or more coded video sequences to be decoded by the video decoder (410); in the same or another embodiment, one coded video sequence at a time, where the decoding of each coded video sequence is independent from other coded video sequences. The coded video sequence may be received from a channel (401), which may be a hardware/software link to a storage device which stores the encoded video data. The receiver (431) may receive the encoded video data with other data, for example, coded audio data and/or ancillary data streams, that may be forwarded to their respective using entities (not depicted). The receiver (431) may separate the coded video sequence from the other data. To combat network jitter, a buffer memory (415) may be coupled in between the receiver (431) and an entropy decoder/parser (420) ("parser (420)" henceforth). In certain applications, the buffer memory (415) is part of the video decoder (410). In others, it can be outside of the video decoder (410) (not depicted). In still others, there can be a buffer memory (not depicted) outside of the video decoder (410), for example to combat network jitter, and in addition another buffer memory (415) inside the video decoder (410), for example to handle playout timing. When

the receiver (431) is receiving data from a store/forward device of sufficient bandwidth and controllability, or from an isosynchronous network, the buffer memory (415) may not be needed, or can be small. For use on best effort packet networks such as the Internet, the buffer memory (415) may be required, can be comparatively large and can be advantageously of adaptive size, and may at least partially be implemented in an operating system or similar elements (not depicted) outside of the video decoder (410).

The video decoder (410) may include the parser (420) to reconstruct symbols (421) from the coded video sequence. Categories of those symbols include information used to manage operation of the video decoder (410), and potentially information to control a rendering device such as a render device (412) (e.g., a display screen) that is not an integral part of the electronic device (430) but can be coupled to the electronic device (430), as was shown in FIG. 4. The control information for the rendering device(s) may be in the form of Supplemental Enhancement Information (SEI messages) or Video Usability Information (VUI) parameter set fragments (not depicted). The parser (420) may parse/entropy-decode the coded video sequence that is received. The coding of the coded video sequence can be in accordance with a video coding technology or standard, and can follow various principles, including variable length coding, Huffman coding, arithmetic coding with or without context sensitivity, and so forth. The parser (420) may extract from the coded video sequence, a set of subgroup parameters for at least one of the subgroups of pixels in the video decoder, based upon at least one parameter corresponding to the group. Subgroups can include Groups of Pictures (GOPs), pictures, tiles, slices, macroblocks, Coding Units (CUs), blocks, Transform Units (TUs), Prediction Units (PUs) and so forth. The parser (420) may also extract from the coded video sequence information such as transform coefficients, quantizer parameter values, MVs, and so forth.

The parser (420) may perform an entropy decoding/parsing operation on the video sequence received from the buffer memory (415), so as to create symbols (421).

Reconstruction of the symbols (421) can involve multiple different units depending on the type of the coded video picture or parts thereof (such as: inter and intra picture, inter and intra block), and other factors. Which units are involved, and how, can be controlled by the subgroup control information that was parsed from the coded video sequence by the parser (420). The flow of such subgroup control information between the parser (420) and the multiple units below is not depicted for clarity.

Beyond the functional blocks already mentioned, the video decoder (410) can be conceptually subdivided into a number of functional units as described below. In a practical implementation operating under commercial constraints, many of these units interact closely with each other and can, at least partly, be integrated into each other. However, for the purpose of describing the disclosed subject matter, the conceptual subdivision into the functional units below is appropriate.

A first unit is the scaler/inverse transform unit (451). The scaler/inverse transform unit (451) receives a quantized transform coefficient as well as control information, including which transform to use, block size, quantization factor, quantization scaling matrices, etc. as symbol(s) (421) from the parser (420). The scaler/inverse transform unit (451) can output blocks comprising sample values that can be input into aggregator (455).

In some cases, the output samples of the scaler/inverse transform (451) can pertain to an intra coded block; that is: a block that is not using predictive information from previously reconstructed pictures, but can use predictive information from previously reconstructed parts of the current picture. Such predictive information can be provided by an intra picture prediction unit (452). In some cases, the intra picture prediction unit (452) generates a block of the same size and shape of the block under reconstruction, using surrounding already reconstructed information fetched from the current picture buffer (458). The current picture buffer (458) buffers, for example, partly reconstructed current picture and/or fully reconstructed current picture. The aggregator (455), in some cases, adds, on a per sample basis, the prediction information that the intra prediction unit (452) has generated to the output sample information as provided by the scaler/inverse transform unit (451).

In other cases, the output samples of the scaler/inverse transform unit (451) can pertain to an inter coded, and potentially motion compensated block. In such a case, a motion compensation prediction unit (453) can access reference picture memory (457) to fetch samples used for prediction. After motion compensating the fetched samples in accordance with the symbols (421) pertaining to the block, these samples can be added by the aggregator (455) to the output of the scaler/inverse transform unit (451) (in this case called the residual samples or residual signal) so as to generate output sample information. The addresses within the reference picture memory (457) from where the motion compensation prediction unit (453) fetches prediction samples can be controlled by MVs, available to the motion compensation prediction unit (453) in the form of symbols (421) that can have, for example X, Y, and reference picture components. Motion compensation also can include interpolation of sample values as fetched from the reference picture memory (457) when sub-sample exact MVs are in use, MV prediction mechanisms, and so forth.

The output samples of the aggregator (455) can be subject to various loop filtering techniques in the loop filter unit (456). Video compression technologies can include in-loop filter technologies that are controlled by parameters included in the coded video sequence (also referred to as coded video bitstream) and made available to the loop filter unit (456) as symbols (421) from the parser (420), but can also be responsive to meta-information obtained during the decoding of previous (in decoding order) parts of the coded picture or coded video sequence, as well as responsive to previously reconstructed and loop-filtered sample values.

The output of the loop filter unit (456) can be a sample stream that can be output to the render device (412) as well as stored in the reference picture memory (457) for use in future inter-picture prediction.

Certain coded pictures, once fully reconstructed, can be used as reference pictures for future prediction. For example, once a coded picture corresponding to a current picture is fully reconstructed and the coded picture has been identified as a reference picture (by, for example, the parser (420)), the current picture buffer (458) can become a part of the reference picture memory (457), and a fresh current picture buffer can be reallocated before commencing the reconstruction of the following coded picture.

The video decoder (410) may perform decoding operations according to a predetermined video compression technology in a standard, such as ITU-T Rec. H.265. The coded video sequence may conform to a syntax specified by the video compression technology or standard being used, in the sense that the coded video sequence adheres to both the

syntax of the video compression technology or standard and the profiles as documented in the video compression technology or standard. Specifically, a profile can select certain tools as the only tools available for use under that profile from all the tools available in the video compression technology or standard. Also necessary for compliance can be that the complexity of the coded video sequence is within bounds as defined by the level of the video compression technology or standard. In some cases, levels restrict the maximum picture size, maximum frame rate, maximum reconstruction sample rate (measured in, for example megasamples per second), maximum reference picture size, and so on. Limits set by levels can, in some cases, be further restricted through Hypothetical Reference Decoder (HRD) specifications and metadata for HRD buffer management signaled in the coded video sequence.

In an embodiment, the receiver (431) may receive additional (redundant) data with the encoded video. The additional data may be included as part of the coded video sequence(s). The additional data may be used by the video decoder (410) to properly decode the data and/or to more accurately reconstruct the original video data. Additional data can be in the form of, for example, temporal, spatial, or signal noise ratio (SNR) enhancement layers, redundant slices, redundant pictures, forward error correction codes, and so on.

FIG. 5 shows a block diagram of a video encoder (503) according to an embodiment of the present disclosure. The video encoder (503) is included in an electronic device (520). The electronic device (520) includes a transmitter (540) (e.g., transmitting circuitry). The video encoder (503) can be used in the place of the video encoder (303) in the FIG. 3 example.

The video encoder (503) may receive video samples from a video source (501) (that is not part of the electronic device (520) in the FIG. 5 example) that may capture video image(s) to be coded by the video encoder (503). In another example, the video source (501) is a part of the electronic device (520).

The video source (501) may provide the source video sequence to be coded by the video encoder (503) in the form of a digital video sample stream that can be of any suitable bit depth (for example: 8 bit, 10 bit, 12 bit, . . .), any colorspace (for example, BT.601 Y CrCb, RGB, . . .), and any suitable sampling structure (for example Y CrCb 4:2:0, Y CrCb 4:4:4). In a media serving system, the video source (501) may be a storage device storing previously prepared video. In a videoconferencing system, the video source (501) may be a camera that captures local image information as a video sequence. Video data may be provided as a plurality of individual pictures that impart motion when viewed in sequence. The pictures themselves may be organized as a spatial array of pixels, wherein each pixel can comprise one or more samples depending on the sampling structure, color space, etc. in use. A person skilled in the art can readily understand the relationship between pixels and samples. The description below focuses on samples.

According to an embodiment, the video encoder (503) may code and compress the pictures of the source video sequence into a coded video sequence (543) in real time or under any other time constraints as required by the application. Enforcing appropriate coding speed is one function of a controller (550). In some embodiments, the controller (550) controls other functional units as described below and is functionally coupled to the other functional units. The coupling is not depicted for clarity. Parameters set by the controller (550) can include rate control related parameters

(picture skip, quantizer, lambda value of rate-distortion optimization techniques, . . .), picture size, group of pictures (GOP) layout, maximum MV allowed reference area, and so forth. The controller (550) can be configured to have other suitable functions that pertain to the video encoder (503) 5 optimized for a certain system design.

In some embodiments, the video encoder (503) is configured to operate in a coding loop. As an oversimplified description, in an example, the coding loop can include a source coder (530) (e.g., responsible for creating symbols, such as a symbol stream, based on an input picture to be coded, and a reference picture(s)), and a (local) decoder (533) embedded in the video encoder (503). The decoder (533) reconstructs the symbols to create the sample data in a similar manner as a (remote) decoder also would create (as any compression between symbols and coded video bit-stream is lossless in the video compression technologies considered in the disclosed subject matter). The reconstructed sample stream (sample data) is input to the reference picture memory (534). As the decoding of a symbol 10 stream leads to bit-exact results independent of decoder location (local or remote), the content in the reference picture memory (534) is also bit exact between the local encoder and remote encoder. In other words, the prediction part of an encoder “sees” as reference picture samples exactly the same sample values as a decoder would “see” when using prediction during decoding. This fundamental principle of reference picture synchronicity (and resulting drift, if synchronicity cannot be maintained, for example because of channel errors) is used in some related arts as well. 15

The operation of the “local” decoder (533) can be the same as of a “remote” decoder, such as the video decoder (410), which has already been described in detail above in conjunction with FIG. 4. Briefly referring also to FIG. 4, however, as symbols are available and encoding/decoding of symbols to a coded video sequence by an entropy coder (545) and the parser (420) can be lossless, the entropy decoding parts of the video decoder (410), including the buffer memory (415) and the parser (420) may not be fully implemented in the local decoder (533). 20

An observation that can be made at this point is that any decoder technology except the parsing/entropy decoding that is present in a decoder also necessarily needs to be present, in substantially identical functional form, in a corresponding encoder. For this reason, the disclosed subject matter focuses on decoder operation. The description of encoder technologies can be abbreviated as they are the inverse of the comprehensively described decoder technologies. Only in certain areas a more detail description is required and provided below. 25

During operation, in some examples, the source coder (530) may perform motion compensated predictive coding, which codes an input picture predictively with reference to one or more previously-coded picture from the video sequence that were designated as “reference pictures”. In this manner, the coding engine (532) codes differences between pixel blocks of an input picture and pixel blocks of reference picture(s) that may be selected as prediction reference(s) to the input picture. 30

The local video decoder (533) may decode coded video data of pictures that may be designated as reference pictures, based on symbols created by the source coder (530). Operations of the coding engine (532) may advantageously be lossy processes. When the coded video data may be decoded at a video decoder (not shown in FIG. 5), the reconstructed video sequence typically may be a replica of the source 35

video sequence with some errors. The local video decoder (533) replicates decoding processes that may be performed by the video decoder on reference pictures and may cause reconstructed reference pictures to be stored in the reference picture cache (534). In this manner, the video encoder (503) may store copies of reconstructed reference pictures locally that have common content as the reconstructed reference pictures that will be obtained by a far-end video decoder (absent transmission errors). 40

The predictor (535) may perform prediction searches for the coding engine (532). That is, for a new picture to be coded, the predictor (535) may search the reference picture memory (534) for sample data (as candidate reference pixel blocks) or certain metadata such as reference picture MVs, block shapes, and so on, that may serve as an appropriate prediction reference for the new pictures. The predictor (535) may operate on a sample block-by-pixel block basis to find appropriate prediction references. In some cases, as determined by search results obtained by the predictor (535), an input picture may have prediction references drawn from multiple reference pictures stored in the reference picture memory (534). 45

The controller (550) may manage coding operations of the source coder (530), including, for example, setting of parameters and subgroup parameters used for encoding the video data. 50

Output of all aforementioned functional units may be subjected to entropy coding in the entropy coder (545). The entropy coder (545) translates the symbols as generated by the various functional units into a coded video sequence, by lossless compressing the symbols according to technologies such as Huffman coding, variable length coding, arithmetic coding, and so forth. 55

The transmitter (540) may buffer the coded video sequence(s) as created by the entropy coder (545) to prepare for transmission via a communication channel (560), which may be a hardware/software link to a storage device which would store the encoded video data. The transmitter (540) may merge coded video data from the video coder (503) with other data to be transmitted, for example, coded audio data and/or ancillary data streams (sources not shown). 60

The controller (550) may manage operation of the video encoder (503). During coding, the controller (550) may assign to each coded picture a certain coded picture type, which may affect the coding techniques that may be applied to the respective picture. For example, pictures often may be assigned as one of the following picture types: 65

An Intra Picture (I picture) may be one that may be coded and decoded without using any other picture in the sequence as a source of prediction. Some video codecs allow for different types of intra pictures, including, for example Independent Decoder Refresh (“IDR”) Pictures. A person skilled in the art is aware of those variants of I pictures and their respective applications and features. 70

A predictive picture (P picture) may be one that may be coded and decoded using intra prediction or inter prediction using at most one MV and reference index to predict the sample values of each block. 75

A bi-directionally predictive picture (B Picture) may be one that may be coded and decoded using intra prediction or inter prediction using at most two MVs and reference indices to predict the sample values of each block. Similarly, multiple-predictive pictures can use more than two reference pictures and associated metadata for the reconstruction of a single block. 80

Source pictures commonly may be subdivided spatially into a plurality of sample blocks (for example, blocks of 85

4×4, 8×8, 4×8, or 16×16 samples each) and coded on a block-by-block basis. Blocks may be coded predictively with reference to other (already coded) blocks as determined by the coding assignment applied to the blocks' respective pictures. For example, blocks of I pictures may be coded non-predictively or they may be coded predictively with reference to already coded blocks of the same picture (spatial prediction or intra prediction). Pixel blocks of P pictures may be coded predictively, via spatial prediction or via temporal prediction with reference to one previously coded reference picture. Blocks of B pictures may be coded predictively, via spatial prediction or via temporal prediction with reference to one or two previously coded reference pictures.

The video encoder (503) may perform coding operations according to a predetermined video coding technology or standard, such as ITU-T Rec. H.265. In its operation, the video encoder (503) may perform various compression operations, including predictive coding operations that exploit temporal and spatial redundancies in the input video sequence. The coded video data, therefore, may conform to a syntax specified by the video coding technology or standard being used.

In an embodiment, the transmitter (540) may transmit additional data with the encoded video. The source coder (530) may include such data as part of the coded video sequence. Additional data may comprise temporal/spatial/SNR enhancement layers, other forms of redundant data such as redundant pictures and slices, SEI messages, VUI parameter set fragments, and so on.

A video may be captured as a plurality of source pictures (video pictures) in a temporal sequence. Intra-picture prediction (often abbreviated to intra prediction) makes use of spatial correlation in a given picture, and inter-picture prediction makes use of the (temporal or other) correlation between the pictures. In an example, a specific picture under encoding/decoding, which is referred to as a current picture, is partitioned into blocks. When a block in the current picture is similar to a reference block in a previously coded and still buffered reference picture in the video, the block in the current picture can be coded by a vector that is referred to as a MV. The MV points to the reference block in the reference picture, and can have a third dimension identifying the reference picture, in case multiple reference pictures are in use.

In some embodiments, a bi-prediction technique can be used in the inter-picture prediction. According to the bi-prediction technique, two reference pictures, such as a first reference picture and a second reference picture that are both prior in decoding order to the current picture in the video (but may be in the past and future, respectively, in display order) are used. A block in the current picture can be coded by a first MV that points to a first reference block in the first reference picture, and a second MV that points to a second reference block in the second reference picture. The block can be predicted by a combination of the first reference block and the second reference block.

Further, a merge mode technique can be used in the inter-picture prediction to improve coding efficiency.

According to some embodiments of the disclosure, predictions, such as inter-picture predictions and intra-picture predictions are performed in the unit of blocks. For example, according to the HEVC standard, a picture in a sequence of video pictures is partitioned into coding tree units (CTU) for compression, the CTUs in a picture have the same size, such as 64×64 pixels, 32×32 pixels, or 16×16 pixels. In general, a CTU includes three coding tree blocks (CTBs), which are

one luma CTB and two chroma CTBs. Each CTU can be recursively quad-tree split into one or multiple coding units (CUs). For example, a CTU of 64×64 pixels can be split into one CU of 64×64 pixels, or 4 CUs of 32×32 pixels, or 16 CUs of 16×16 pixels. In an example, each CU is analyzed to determine a prediction type for the CU, such as an inter prediction type or an intra prediction type. The CU is split into one or more prediction units (PUs) depending on the temporal and/or spatial predictability. Generally, each PU includes a luma prediction block (PB), and two chroma PBs. In an embodiment, a prediction operation in coding (encoding/decoding) is performed in the unit of a prediction block. Using a luma prediction block as an example of a prediction block, the prediction block includes a matrix of values (e.g., luma values) for pixels, such as 8×8 pixels, 16×16 pixels, 8×16 pixels, 16×8 pixels, and the like.

FIG. 6 shows a diagram of a video encoder (603) according to another embodiment of the disclosure. The video encoder (603) is configured to receive a processing block (e.g., a prediction block) of sample values within a current video picture in a sequence of video pictures, and encode the processing block into a coded picture that is part of a coded video sequence. In an example, the video encoder (603) is used in the place of the video encoder (303) in the FIG. 3 example.

In an HEVC example, the video encoder (603) receives a matrix of sample values for a processing block, such as a prediction block of 8×8 samples, and the like. The video encoder (603) determines whether the processing block is best coded using intra mode, inter mode, or bi-prediction mode using, for example, rate-distortion optimization. When the processing block is to be coded in intra mode, the video encoder (603) may use an intra prediction technique to encode the processing block into the coded picture; and when the processing block is to be coded in inter mode or bi-prediction mode, the video encoder (603) may use an inter prediction or bi-prediction technique, respectively, to encode the processing block into the coded picture. In certain video coding technologies, merge mode can be an inter picture prediction submode where the MV is derived from one or more MV predictors without the benefit of a coded MV component outside the predictors. In certain other video coding technologies, a MV component applicable to the subject block may be present. In an example, the video encoder (603) includes other components, such as a mode decision module (not shown) to determine the mode of the processing blocks.

In the FIG. 6 example, the video encoder (603) includes the inter encoder (630), an intra encoder (622), a residue calculator (623), a switch (626), a residue encoder (624), a general controller (621), and an entropy encoder (625) coupled together as shown in FIG. 6.

The inter encoder (630) is configured to receive the samples of the current block (e.g., a processing block), compare the block to one or more reference blocks in reference pictures (e.g., blocks in previous pictures and later pictures), generate inter prediction information (e.g., description of redundant information according to inter encoding technique, MVs, merge mode information), and calculate inter prediction results (e.g., prediction block) based on the inter prediction information using any suitable technique. In some examples, the reference pictures are decoded reference pictures that are decoded based on the encoded video information.

The intra encoder (622) is configured to receive the samples of the current block (e.g., a processing block), in some cases compare the block to blocks already coded in the

same picture, generate quantized coefficients after transform, and in some cases also intra prediction information (e.g., an intra prediction direction information according to one or more intra encoding techniques). In an example, the intra encoder (622) also calculates intra prediction results (e.g., prediction block) based on the intra prediction information and reference blocks in the same picture.

The general controller (621) is configured to determine general control data and control other components of the video encoder (603) based on the general control data. In an example, the general controller (621) determines the mode of the block, and provides a control signal to the switch (626) based on the mode. For example, when the mode is the intra mode, the general controller (621) controls the switch (626) to select the intra mode result for use by the residue calculator (623), and controls the entropy encoder (625) to select the intra prediction information and include the intra prediction information in the bitstream; and when the mode is the inter mode, the general controller (621) controls the switch (626) to select the inter prediction result for use by the residue calculator (623), and controls the entropy encoder (625) to select the inter prediction information and include the inter prediction information in the bitstream.

The residue calculator (623) is configured to calculate a difference (residue data) between the received block and prediction results selected from the intra encoder (622) or the inter encoder (630). The residue encoder (624) is configured to operate based on the residue data to encode the residue data to generate the transform coefficients. In an example, the residue encoder (624) is configured to convert the residue data from a spatial domain to a frequency domain, and generate the transform coefficients. The transform coefficients are then subject to quantization processing to obtain quantized transform coefficients. In various embodiments, the video encoder (603) also includes a residue decoder (628). The residue decoder (628) is configured to perform inverse-transform, and generate the decoded residue data. The decoded residue data can be suitably used by the intra encoder (622) and the inter encoder (630). For example, the inter encoder (630) can generate decoded blocks based on the decoded residue data and inter prediction information, and the intra encoder (622) can generate decoded blocks based on the decoded residue data and the intra prediction information. The decoded blocks are suitably processed to generate decoded pictures and the decoded pictures can be buffered in a memory circuit (not shown) and used as reference pictures in some examples.

The entropy encoder (625) is configured to format the bitstream to include the encoded block. The entropy encoder (625) is configured to include various information according to a suitable standard such as HEVC. In an example, the entropy encoder (625) is configured to include the general control data, the selected prediction information (e.g., intra prediction information or inter prediction information), the residue information, and other suitable information in the bitstream. Note that, according to the disclosed subject matter, when coding a block in the merge submode of either inter mode or bi-prediction mode, there is no residue information.

FIG. 7 shows a diagram of a video decoder (710) according to another embodiment of the disclosure. The video decoder (710) is configured to receive coded pictures that are part of a coded video sequence, and decode the coded pictures to generate reconstructed pictures. In an example, the video decoder (710) is used in the place of the video decoder (310) in the FIG. 3 example.

In the FIG. 7 example, the video decoder (710) includes an entropy decoder (771), an inter decoder (780), a residue decoder (773), a reconstruction module (774), and an intra decoder (772) coupled together as shown in FIG. 7.

The entropy decoder (771) can be configured to reconstruct, from the coded picture, certain symbols that represent the syntax elements of which the coded picture is made up. Such symbols can include, for example, the mode in which a block is coded (such as, for example, intra mode, inter mode, bi-predicted mode, the latter two in merge submode or another submode), prediction information (such as, for example, intra prediction information or inter prediction information) that can identify certain sample or metadata that is used for prediction by the intra decoder (772) or the inter decoder (780), respectively, residual information in the form of, for example, quantized transform coefficients, and the like. In an example, when the prediction mode is inter or bi-predicted mode, the inter prediction information is provided to the inter decoder (780); and when the prediction type is the intra prediction type, the intra prediction information is provided to the intra decoder (772). The residual information can be subject to inverse quantization and is provided to the residue decoder (773).

The inter decoder (780) is configured to receive the inter prediction information, and generate inter prediction results based on the inter prediction information.

The intra decoder (772) is configured to receive the intra prediction information, and generate prediction results based on the intra prediction information.

The residue decoder (773) is configured to perform inverse quantization to extract de-quantized transform coefficients, and process the de-quantized transform coefficients to convert the residual from the frequency domain to the spatial domain. The residue decoder (773) may also require certain control information (to include the Quantizer Parameter (QP)), and that information may be provided by the entropy decoder (771) (data path not depicted as this may be low volume control information only).

The reconstruction module (774) is configured to combine, in the spatial domain, the residual as output by the residue decoder (773) and the prediction results (as output by the inter or intra prediction modules as the case may be) to form a reconstructed block, that may be part of the reconstructed picture, which in turn may be part of the reconstructed video. It is noted that other suitable operations, such as a deblocking operation and the like, can be performed to improve the visual quality.

It is noted that the video encoders (303), (503), and (603), and the video decoders (310), (410), and (710) can be implemented using any suitable technique. In an embodiment, the video encoders (303), (503), and (603), and the video decoders (310), (410), and (710) can be implemented using one or more integrated circuits. In another embodiment, the video encoders (303), (503), and (603), and the video decoders (310), (410), and (710) can be implemented using one or more processors that execute software instructions.

II. Video Coding For Machine

Video or images can be consumed by human beings for variety of usages such as entertainment, education, and the like. Thus, video coding or image coding often utilizes characteristics of the human visual system for a better compression efficiency while maintaining a good subjective quality.

In recent years, with the rise of machine learning applications, along with being equipped with an abundance of sensors, many intelligent platforms have utilized video or

images for machine vision tasks such as object detection, segmentation, or tracking. How to encode video or images for consumption of the machine vision tasks can be an interesting and challenging problem. This has led to an introduction of video coding for machines (VCM) studies. To achieve this goal, the international standard group MPEG created an Ad-Hoc group, "Video coding for machine (VCM)" to standardize the related techniques for a better interoperability among difference devices.

FIG. 8 shows an exemplary VCM architecture according to an embodiment of the disclosure. It is noted that an output of a video decoding module in FIG. 8 is mainly for machine consumption, i.e., machine vision. However, in some cases, the output of the video decoding module can also be used for human vision, as indicated in FIG. 8. In addition, an interface for neural network (NN) module can be included in the exemplary VCM architecture.

A video coding for machine system, from a client (or a decoder) end, can typically perform video decoding to obtain a video sequence in a sample domain first. Then, one or more machine tasks can be performed to understand video content of the video sequence. In some cases, an output of the one or more machine tasks can be labeling information of targeted object(s), for example, in a form of rectangular bounding boxes inside the decoded image or video sequence.

III. Versatile Video Coding

VVC was recently jointly developed by two international standard organizations, i.e., ITU and ISO/IEC. One version of VVC was finalized in July 2020 and is one of the state-of-the-art video coding standards.

FIG. 9 shows an exemplary VVC encoder according to an embodiment of the disclosure. The VVC encoder includes a transform and quantization (TR+Q) module, an entropy coding module, a motion estimation (ME) module, and a decoder. The decoder includes an inverse transform and inverse quantization (iTR+iQ) module, an intra prediction module, an inter prediction module, a combined prediction module, a deblocking filter (Deblk) module, a sample adaptive offset filter (SAO) module, an adaptive loop filter (ALF) module, and a buffer. The encoder receives a coding block (CB) of an input picture and outputs a coded bitstream and a reconstructed picture. Both the transform and quantization module and the inverse transform and inverse quantization module process transform blocks (TB). Both the intra prediction module and the intra prediction module process PB. The combined prediction module combines inter prediction and intra prediction.

For intra prediction, a variety of coding tools can be included, for example, cross-component linear mode (CCLM), multiple reference line prediction (MRLP), intra sub-partitioning (ISP), matrix based intra prediction (MIP), and/or the like.

For inter prediction, a set of coding tools can be included, for example, affine motion model (AFFINE), subblock-based temporal merging candidates (SbTMC), adaptive motion vector resolution (AMVR), geometric partition mode (GPM), combined intra/inter prediction (CIIP), merge mode with motion vector difference (MMVD), bi-predictive with CU weights (BCW), decoder motion vector refinement (DMVR), bi-directional optical flow (BDOF), prediction refinement using optical flow (PROF), and/or the like.

Other tools, such as transform, quantization, and in-loop filters can be included, for example, dependent quantization (DQ), multiple transform set (MTS), low frequency non-separable transform (LFNST), luma mapping and chroma scaling (LMCS), cross-component adaptive loop filter

(CCALF), adaptive loop filter (ALF), sample adaptive offset filter (SAO), deblocking filter (Deblk), and/or the like.

In a video coding system, some useful information that is not necessary for correctly decoding the video bitstream, can be delivered, such as by one or more supplementary enhancement information (SEI) messages. A decoder can, for example, disregard such information. Alternatively, the decoder can decode the SEI messages for a usage after decoding the video sequence. In one example, in a virtual reality (VR) 360-degree video streaming, an omni-directional video can be packed into 2-D traditional video for compression and transmission. After decoding the 2-D video, the system may need to handle the packed 2-D video in several processing steps, before the video can be viewed in a 3-D format. For example, converting the packed 2-D video back to a spherical domain according to the packing format such as equi-rectangular projection (ERP), cube-map projection (CMP), and the like), displaying the spherical signal with an assumed center view which contains the most relevant information of the video content. These steps are not related to correctly decoding the video itself but are useful for displaying the video content. Information such as the packing method (ERP, CMP, and the like), or information of a suggested viewing center of the 360-degree video, can be sent via the SEI messages. By receiving the information, the VR client system can handle the decoded video content more efficiently.

IV. Signaling of Labeling Information of Objects

In some related video coding standards, such as H.264, HEVC or the recently finalized VVC standards, an input video signal (or sequence) is treated as a waveform, without an understanding of a video content meaning, such as how many people or objects are in the video, how they move around, and the like. On the other hand, machine vision tasks such as object detection, segmentation, or tracking, can be designed to understand these types of information from the video sequence. After decoding the video sequence, a client (or decoder) end can perform the machine tasks to obtain labeling information of targeted object(s) in the image or video sequence.

Performing machine tasks at the client (or decoder) side can incur not only a consumption of computation, but also a consumption of time to obtain the results of the machine tasks. In some systems, those costs (e.g., consumptions of computation and time) are not desirable.

This disclosure includes methods of sending detection or recognition information, such as labeling results, to a client (or decoder) end. It is noted that the methods included in this disclosure are not limited to VVC standards. The principles of the methods can also be applied to other video coding standards, such as H.264, HEVC, or AV1 which is developed by Alliance for Open Media.

According to aspects of the disclosure, labeling information such as one or more bounding boxes, classes, and/or indexes of objects for machine vision targets can be determined at an encoder side, rather than being determined after decoding a coded video bitstream at a decoder side. The determined labeling information can be coded and sent to the decoder and/or client side in various manners. After decoding the labeling information, the labeling results can be directly applied to the decoded image or video sequence, without further performing the machine tasks to understand the video content. In an example, each of the one or more bounding boxes can be applied to a different target object in the decoded image. In another example, a class and/or an index of each target object can be displayed in the decoded

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image. In this way, the machine tasks can be performed independent of the resolution of the picture(s).

In an embodiment, the encoder system can apply either a similar machine vision task module such as the module illustrated in FIG. 9 to acquire the related labeling information or use a different one. For example, an output of the machine vision tasks can be in a form of a series of bounding boxes for each picture. Each bounding box can be represented by location and size information of the respective bounding box.

According to aspects of the disclosure, the labeling information such as information of bounding boxes of objects in each picture can be delivered by some side information messages in a coded video bitstream. For example, the labeling information can be sent through SEI messages in the coded video bitstream. In another embodiment, the labeling information can also be provided separately, such as in a separate file or a plug-in file associated with the coded video bitstream.

In some embodiments, the labeling information includes location information of one or more bounding boxes in each picture. Various reference points and/or size information can be used to indicate the location of the bounding boxes. In an embodiment, the location information of a bounding box can include a top left position of the bounding box and a size of the bounding box.

Table 1 shows an exemplary syntax table that can be used for signaling labeling information including location information of bounding boxes in a current picture.

TABLE 1

Descriptor	
bounding_boxes_sei() {	
...	
num_boxes	u(N)
for(i = 0; i < num_boxes; i++) {	
box_loc_x[i]	u(16)
box_loc_y[i]	u(16)
box_width[i]	u(16)
box_height[i]	u(16)
}	
}	

In Table 1, the syntax element num_boxes indicates a total number of the bounding boxes inside the current picture. The syntax element box_loc_x[i] indicates a horizontal position of a top left corner of the i-th bounding box, relative to the top left corner of the current picture, in luma samples. The syntax element box_loc_y[i] indicates a vertical position of the top left corner of the i-th bounding box, relative to the top left corner of the current picture, in luma samples. The syntax element box_width[i] indicates a width of the i-th bounding box in luma samples. The syntax element box_height[i] indicates a height of the i-th bounding box in luma samples.

Note that a representation of a bounding box may not be limited by the above format (top left position+size). For example, a top-left position+a bottom-right position can also describe the bounding box.

Note also that a location of a bounding box can be represented by any fixed position of the box. The top left corner is one example, and other locations may also be utilized. For example, a center position of a bounding box and a width and a height of the bounding box can be used to represent the bounding box.

In one embodiment, the labeling information can include category information of each object in a current picture. The

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category information indicates a category to which an object in a bounding box belongs. For example, a category can be a person, a car, a plane, and/or the like. Table 2 shows an exemplary syntax table that can be used to signal labeling information including location information of bounding boxes in a current picture and category information of objects in the current picture.

TABLE 2

Descriptor	
bounding_boxes_sei() {	
...	
num_boxes	u(N)
for(i = 0; i < num_boxes; i++) {	
category_id	u(8)
box_loc_x[i]	u(6)
box_loc_y[i]	u(16)
box_width[i]	u(16)
box_height[i]	u(16)
}	
}	

In Table 2, the syntax element num_boxes indicates a total number of the bounding boxes inside the current picture. The syntax element category_id indicates a category to which an object in the i-th bounding belongs. The syntax element box_loc_x[i] indicates a horizontal position of a top left corner of the i-th bounding box, relative to the top left corner of a current picture, in luma samples. The syntax element box_loc_y[i] indicates a vertical position of the top left corner of the i-th bounding box, relative to the top left corner of the current picture, in luma samples. The syntax element box_width[i] indicates a width of the i-th bounding box in luma samples. The syntax element box_height[i] indicates a height of the i-th bounding box in luma samples.

In one embodiment, besides the category information, the labeling information can include identification information of one or more objects in a current picture. In one example, the labeling information can include identification information for each object in a current picture. The identification information can be used to identify an object in a video sequence. For example, in object tracking, the identification information can be used to represent the same object in the video sequence. Table 3 shows a syntax table used for signaling labeling information including location information of bounding boxes in a current picture and category information and identification information of objects in the current picture.

TABLE 3

Descriptor	
bounding_boxes_sei() {	
...	
num_boxes	u(N)
for(i = 0; i < num_boxes; i++) {	
category_id	u(8)
Instance_id	u(8)
box_loc_x[i]	u(16)
box_loc_y[i]	u(16)
box_width[i]	u(16)
box_height[i]	u(16)
}	
}	

In Table 3, the syntax element num_boxes indicates a total number of the bounding boxes inside the current picture. The syntax element category_id indicates a category to which an object in the i-th bounding belongs. The syntax

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element instance_id denotes an identification number for the object in the i-th bounding. The syntax element box_loc_x[i] indicates a horizontal position of a top left corner of the i-th bounding box, relative to the top left corner of the current picture, in luma samples. The syntax element box_loc_y[i] indicates a vertical position of the top left corner of the i-th bounding box, relative to the top left corner of the current picture, in luma samples. The syntax element box_width[i] indicates a width of the i-th bounding box in luma samples. The syntax element box_height[i] indicates a height of the i-th bounding box in luma samples.

In some machine vision tasks such as object tracking related tasks, each object may appear in different pictures. To facilitate the signaling of the same bounding box across different pictures, the same bounding box id can be used to represent the same object. In a subsequent picture(s), an absolute position and a size of the bounding box can be signaled. In another embodiment, a relative change to its previous value in a previous picture can be used instead. Table 4 shows an exemplary syntax table to can be used to signal the labeling information including location offset information of bounding boxes in a current picture. In Table 4, a bounding box in the current picture is described by signaling a top left corner position and a bottom right corner position of the bounding box.

TABLE 4

Descriptor	
<pre> bounding_boxes_sei() { ... num_boxes for(i = 0; i < num_boxes; i++) { sign_tl_x[i] sign_tl_y[i] sign_br_x[i] sign_br_y[i] delta_box_tl_loc_x[i] delta_box_tl_loc_y[i] delta_box_br_loc_x[i] delta_box_br_loc_y[i] } }</pre>	
	u(N)
	u(1)
	u(1)
	u(1)
	u(1)
	u(16)
	u(16)
	u(16)
	u(16)

In Table 4, the syntax element num_boxes indicates a total number of bounding boxes inside the current picture. The syntax element sign_tl_x[i] indicates a sign of a horizontal top left corner position difference of the i-th bounding box relative to the same bounding box in a previous picture, in luma samples. The syntax element sign_tl_y[i] indicates a sign of a vertical top left corner position difference of the i-th bounding box relative to the same bounding box in the previous picture, in luma samples. The syntax element sign_br_x[i] indicates a sign of a horizontal bottom right corner position difference of the i-th bounding box relative to the same bounding box in the previous picture, in luma samples. The syntax element sign_br_y[i] indicates a sign of a vertical bottom right corner position difference of the i-th bounding box relative to the same bounding box in the previous picture, in luma samples. The syntax element delta_box_tl_loc_x[i] indicates an absolute value of a horizontal top left corner position difference of the i-th bounding box relative to the same bounding box in the previous picture, in luma samples. The syntax element delta_box_tl_loc_y[i] indicates an absolute value of a vertical top left corner position difference of the i-th bounding box relative to the same bounding box in the previous picture, in luma samples. The syntax element delta_box_br_loc_x[i] indicates an absolute value of a horizontal bottom right corner

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position difference of the i-th bounding box relative to the same bounding box in the previous picture, in luma samples. The syntax element delta_box_br_loc_y[i] indicates an absolute value of a vertical bottom right corner position difference of the i-th bounding box relative to the same bounding box in the previous picture, in luma samples.

The syntax elements sign_tl_x[i], sign_tl_y[i], sign_br_x[i], sign_br_y[i] can be set equal to 0, when there is no previous picture prior to the current picture. Alternatively, these syntax elements can be conditionally signaled only when there is a previous picture relative to the current picture in a decoding order or in a display order. When not signaled, these syntax elements can be inferred to be 0.

A variable PrevTopLeftBoxX[i] indicates the horizontal top left corner position of the i-th bounding box in the previous picture prior to the current picture in the decoding order or the display order. A variable TopLeftBoxX[i] indicates the horizontal top left corner position of the i-th bounding box in the current picture.

$TopLeftBoxX[i] = PrevTopLeftBoxX[i] + sign_tl_x[i] * delta_box_tl_loc_x[i]$.

A variable PrevTopLeftBoxY[i] indicates the vertical top left corner position of the i-th bounding box in the previous picture prior to the current picture in the decoding order or the display order. A variable TopLeftBoxY[i] indicates the vertical top left corner position of the i-th bounding box in the current picture.

$TopLeftBoxY[i] = PrevTopLeftBoxY[i] + sign_tl_y[i] * delta_box_tl_loc_y[i]$.

A variable PrevBotRightBoxX[i] indicates the horizontal bottom-right corner position of the i-th bounding box in the previous picture prior to the current picture in the decoding order or the display order. A variable BotRightBoxX[i] indicates the horizontal bottom-right corner position of the i-th bounding box in the current picture.

$BotRightBoxX[i] = PrevBotRightBoxX[i] + sign_br_x[i] * delta_box_br_loc_x[i]$.

A variable PrevBotRightBoxY[i] indicates the vertical bottom-right corner position of the i-th bounding box in the previous picture prior to the current picture in the decoding order or the display order. A variable BotRightBoxY[i] indicates the vertical bottom-right corner position of the i-th bounding box in the current picture.

$BotRightBoxY[i] = PrevBotRightBoxY[i] + sign_br_y[i] * delta_box_br_loc_y[i]$.

The variables PrevTopLeftBoxX[i], PrevTopLeftBoxY[i], PrevBotRightBoxX[i], PrevBotRightBoxY[i] can be initialized as 0 when there is no previous picture prior to the current picture.

The variables PrevTopLeftBoxX[i], PrevTopLeftBoxY[i], PrevBotRightBoxX[i], PrevBotRightBoxY[i] can be set equal to TopLeftBoxX[i], TopLeftBoxY[i], BotRightBoxX[i], BotRightBoxY[i], respectively, after a completion of decoding or displaying the current picture.

In an embodiment, a signaled syntax structure is similar to the above syntax tables, but a bounding box in a current picture can be described by signaling its top left corner position plus its width and height. In this case, the delta values to be signaled can be the delta values of the top left corner position of the i-th bounding box, the delta values of the width and height of the i-th bounding box, and the respective sign values.

In an embodiment, the category information (e.g., category_id) and/or identification information (e.g., instance_id) of an object can be included in Table 4.

According to aspects of the disclosure, the top left and bottom right positions of a bounding box can be placed

inside a current picture. However, in some cases, when an object no longer appears in the current picture, the bounding box should not be displayed in the current picture anymore. In one embodiment, for the *i*-th bounding box not to be shown in the current picture, a predetermined position value can be used, such as a position value outside the picture boundary. For example, if the picture size is 1920×1080, the top left position and bottom right positions of the bounding box can be set as 2000 (larger than the picture width and height) to indicate that the object does not exist in the current picture and the bounding box of the object does not need to be displayed.

According to aspects of the disclosure, other than being included in metadata information (e.g., SEI messages) of a coded video bitstream, the labeling information can be included in a metadata file (e.g., a plug-in file) separated from the coded video bitstream. The labeling information can be regarded as the metadata information and delivered to a decoder and/or client side via a system layer approach, such as real-time transport protocol (RTP), ISO base media format file, and dynamic adaptive streaming over hypertext transport protocol (DASH).

The client (or decoder) side, when needed, can request such information from the system layer, to enable the labeling information in the decoded picture(s).

In an embodiment, the client (or decoder) side can send a request message to the encoder side to request the labeling information from the system layer.

In an embodiment, the client (or decoder) side can receive the labeling information from the encoder side without sending the request message.

In an embodiment, the client can send a request message to the decoder to enable the labeling information in the decoded picture(s).

In an embodiment, the decoder can enable the labeling information in the decoder picture(s) without receiving the request message from the client.

V. Flowchart

FIG. 10 shows a flow chart outlining an exemplary process (1000) according to an embodiment of the disclosure. In various embodiments, the process (1000) is executed by processing circuitry, such as the processing circuitry in the terminal devices (210), (220), (230) and (240), the processing circuitry that performs functions of the video encoder (303), the processing circuitry that performs functions of the video decoder (310), the processing circuitry that performs functions of the video decoder (410), the processing circuitry that performs functions of the intra prediction module (452), the processing circuitry that performs functions of the video encoder (503), the processing circuitry that performs functions of the predictor (535), the processing circuitry that performs functions of the intra encoder (622), the processing circuitry that performs functions of the intra decoder (772), and the like. In some embodiments, the process (1000) is implemented in software instructions, thus when the processing circuitry executes the software instructions, the processing circuitry performs the process (1000).

The process (1000) may generally start at step (S1010), where the process (1000) receives metadata associated with a coded video bitstream. The metadata includes labeling information of one or more objects detected in a first picture that is coded in the coded video bitstream. Then, the process (1000) proceeds to step (S1020).

At step (S1020), the process (1000) decodes the labeling information of the one or more objects in the first picture that

is coded in the coded video bitstream. Then, the process (1000) proceeds to step (S1030).

At step (S1030), the process (1000) applies the labeling information to the one or more objects in the first picture. Then, the process (1000) terminates.

In an example, a bounding box can be applied to a detected object in the first picture. In another example, a category to which the detected object belongs can be displayed in the first picture. In another example, an identification number of the detected object can be displayed in the first picture.

In one embodiment, the metadata is included in an SEI message in the coded video bitstream.

In one embodiment, the metadata is included in a file that is separate from the coded video bitstream.

In one embodiment, the labeling information indicates a total number of bounding boxes in the first picture and includes location information and size information of each bounding box, each bounding box being associated with one of the one or more objects in the first picture.

In one embodiment, the labeling information includes category information that indicates a category for each of the one or more objects.

In one embodiment, the labeling information includes identification information that identifies each of the one or more objects in a video sequence.

In one embodiment, the location information of one of the bounding boxes includes a location offset of the one of the bounding boxes between the first picture and a second picture that is coded in the video bitstream.

In one embodiment, the location information of one of the bounding boxes indicates a location outside the first picture for the one of the bounding boxes based on an object associated with the one of the bounding boxes not existing in the first picture.

In one embodiment, the process (1000) sends a request to receive the metadata associated with the coded video bitstream.

VI. Computer System

The techniques described above, can be implemented as computer software using computer-readable instructions and physically stored in one or more computer-readable media. For example, FIG. 11 shows a computer system (1100) suitable for implementing certain embodiments of the disclosed subject matter.

The computer software can be coded using any suitable machine code or computer language, that may be subject to assembly, compilation, linking, or like mechanisms to create code comprising instructions that can be executed directly, or through interpretation, micro-code execution, and the like, by one or more computer central processing units (CPUs), Graphics Processing Units (GPUs), and the like.

The instructions can be executed on various types of computers or components thereof, including, for example, personal computers, tablet computers, servers, smartphones, gaming devices, internet of things devices, and the like.

The components shown in FIG. 11 for computer system (1100) are exemplary in nature and are not intended to suggest any limitation as to the scope of use or functionality of the computer software implementing embodiments of the present disclosure. Neither should the configuration of components be interpreted as having any dependency or requirement relating to any one or combination of components illustrated in the exemplary embodiment of a computer system (1100).

Computer system (1100) may include certain human interface input devices. Such a human interface input device

may be responsive to input by one or more human users through, for example, tactile input (such as: keystrokes, swipes, data glove movements), audio input (such as: voice, clapping), visual input (such as: gestures), olfactory input (not depicted). The human interface devices can also be used to capture certain media not necessarily directly related to conscious input by a human, such as audio (such as: speech, music, ambient sound), images (such as: scanned images, photographic images obtain from a still image camera), video (such as two-dimensional video, three-dimensional video including stereoscopic video).

Input human interface devices may include one or more of (only one of each depicted): keyboard (1101), mouse (1102), trackpad (1103), touch screen (1110), data-glove (not shown), joystick (1105), microphone (1106), scanner (1107), and camera (1108).

Computer system (1100) may also include certain human interface output devices. Such human interface output devices may be stimulating the senses of one or more human users through, for example, tactile output, sound, light, and smell/taste. Such human interface output devices may include tactile output devices (for example tactile feedback by the touch-screen (1110), data-glove (not shown), or joystick (1105), but there can also be tactile feedback devices that do not serve as input devices), audio output devices (such as: speakers (1109), headphones (not depicted)), visual output devices (such as screens (1110) to include CRT screens, LCD screens, plasma screens, OLED screens, each with or without touch-screen input capability, each with or without tactile feedback capability—some of which may be capable to output two dimensional visual output or more than three dimensional output through means such as stereographic output; virtual-reality glasses (not depicted), holographic displays and smoke tanks (not depicted)), and printers (not depicted). These visual output devices (such as screens (1110)) can be connected to a system bus (1148) through a graphics adapter (1150).

Computer system (1100) can also include human accessible storage devices and their associated media such as optical media including CD/DVD ROM/RW (1120) with CD/DVD or the like media (1121), thumb-drive (1122), removable hard drive or solid state drive (1123), legacy magnetic media such as tape and floppy disc (not depicted), specialized ROM/ASIC/PLD based devices such as security dongles (not depicted), and the like.

Those skilled in the art should also understand that term “computer readable media” as used in connection with the presently disclosed subject matter does not encompass transmission media, carrier waves, or other transitory signals.

Computer system (1100) can also include a network interface (1154) to one or more communication networks (1155). The one or more communication networks (1155) can for example be wireless, wireline, optical. The one or more communication networks (1155) can further be local, wide-area, metropolitan, vehicular and industrial, real-time, delay-tolerant, and so on. Examples of the one or more communication networks (1155) include local area networks such as Ethernet, wireless LANs, cellular networks to include GSM, 3G, 4G, 5G, LTE and the like, TV wireline or wireless wide area digital networks to include cable TV, satellite TV, and terrestrial broadcast TV, vehicular and industrial to include CANBus, and so forth. Certain networks commonly require external network interface adapters that attached to certain general purpose data ports or peripheral buses (1149) (such as, for example USB ports of the computer system (1100)); others are commonly integrated into the core of the computer system (1100) by

attachment to a system bus as described below (for example Ethernet interface into a PC computer system or cellular network interface into a smartphone computer system). Using any of these networks, computer system (1100) can communicate with other entities. Such communication can be uni-directional, receive only (for example, broadcast TV), uni-directional send-only (for example CANbus to certain CANbus devices), or bi-directional, for example to other computer systems using local or wide area digital networks. Certain protocols and protocol stacks can be used on each of those networks and network interfaces as described above.

Aforementioned human interface devices, human-accessible storage devices, and network interfaces can be attached to a core (1140) of the computer system (1100).

The core (1140) can include one or more Central Processing Units (CPU) (1141), Graphics Processing Units (GPU) (1142), specialized programmable processing units in the form of Field Programmable Gate Areas (FPGA) (1143), hardware accelerators for certain tasks (1144), graphics adapters (1150), and so forth. These devices, along with Read-only memory (ROM) (1145), Random-access memory (1146), internal mass storage (1147) such as internal non-user accessible hard drives, SSDs, and the like, may be connected through the system bus (1148). In some computer systems, the system bus (1148) can be accessible in the form of one or more physical plugs to enable extensions by additional CPUs, GPU, and the like. The peripheral devices can be attached either directly to the core's system bus (1148), or through a peripheral bus (1149). In an example, the screen (1110) can be connected to the graphics adapter (1150). Architectures for a peripheral bus include PCI, USB, and the like.

CPUs (1141), GPUs (1142), FPGAs (1143), and accelerators (1144) can execute certain instructions that, in combination, can make up the aforementioned computer code. That computer code can be stored in ROM (1145) or RAM (1146). Transitional data can be also be stored in RAM (1146), whereas permanent data can be stored for example, in the internal mass storage (1147). Fast storage and retrieve to any of the memory devices can be enabled through the use of cache memory, that can be closely associated with one or more CPU (1141), GPU (1142), mass storage (1147), ROM (1145), RAM (1146), and the like.

The computer readable media can have computer code thereon for performing various computer-implemented operations. The media and computer code can be those specially designed and constructed for the purposes of the present disclosure, or they can be of the kind well known and available to those having skill in the computer software arts.

As an example and not by way of limitation, the computer system having architecture (1100), and specifically the core (1140) can provide functionality as a result of processor(s) (including CPUs, GPUs, FPGA, accelerators, and the like) executing software embodied in one or more tangible, computer-readable media. Such computer-readable media can be media associated with user-accessible mass storage as introduced above, as well as certain storage of the core (1140) that are of non-transitory nature, such as core-internal mass storage (1147) or ROM (1145). The software implementing various embodiments of the present disclosure can be stored in such devices and executed by core (1140). A computer-readable medium can include one or more memory devices or chips, according to particular needs. The software can cause the core (1140) and specifically the processors therein (including CPU, GPU, FPGA, and the like) to execute particular processes or particular parts of

particular processes described herein, including defining data structures stored in RAM (1146) and modifying such data structures according to the processes defined by the software. In addition or as an alternative, the computer system can provide functionality as a result of logic hard-wired or otherwise embodied in a circuit (for example: accelerator (1144)), which can operate in place of or together with software to execute particular processes or particular parts of particular processes described herein. Reference to software can encompass logic, and vice versa, where appropriate. Reference to a computer-readable media can encompass a circuit (such as an integrated circuit (IC)) storing software for execution, a circuit embodying logic for execution, or both, where appropriate. The present disclosure encompasses any suitable combination of hardware and software.

While this disclosure has described several exemplary embodiments, there are alterations, permutations, and various substitute equivalents, which fall within the scope of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise numerous systems and methods which, although not explicitly shown or described herein, embody the principles of the disclosure and are thus within the spirit and scope thereof.

APPENDIX A: ACRONYMS

ALF: Adaptive Loop Filter
 AMVP: Advanced Motion Vector Prediction
 APS: Adaptation Parameter Set
 ASIC: Application-Specific Integrated Circuit
 ATMVP: Alternative/Advanced Temporal Motion Vector Prediction
 AV1: AOMedia Video 1
 AV2: AOMedia Video 2
 BMS: Benchmark Set
 BV: Block Vector
 CANB us: Controller Area Network Bus
 CB: Coding Block
 CC-ALF: Cross-Component Adaptive Loop Filter
 CD: Compact Disc
 CDEF: Constrained Directional Enhancement Filter
 CPR: Current Picture Referencing
 CPU: Central Processing Unit
 CRT: Cathode Ray Tube
 CTB: Coding Tree Block
 CTU: Coding Tree Unit
 CU: Coding Unit
 DPB: Decoder Picture Buffer
 DPCM: Differential Pulse-Code Modulation
 DPS: Decoding Parameter Set
 DVD: Digital Video Disc
 FPGA: Field Programmable Gate Area
 JCCR: Joint CbCr Residual Coding
 JVET: Joint Video Exploration Team
 GOP: Groups of Pictures
 GPU: Graphics Processing Unit
 GSM: Global System for Mobile communications
 HDR: High Dynamic Range
 HEVC: High Efficiency Video Coding
 HRD: Hypothetical Reference Decoder
 IBC: Intra Block Copy
 IC: Integrated Circuit
 ISP: Intra Sub-Partitions
 JEM: Joint Exploration Model
 LAN: Local Area Network
 LCD: Liquid-Crystal Display

LR: Loop Restoration Filter
 LRU: Loop Restoration Unit
 LTE: Long-Term Evolution
 MPM: Most Probable Mode
 MV: Motion Vector
 OLED: Organic Light-Emitting Diode
 PBs: Prediction Blocks
 PCI: Peripheral Component Interconnect
 PDPC: Position Dependent Prediction Combination
 PLD: Programmable Logic Device
 PPS: Picture Parameter Set
 PU: Prediction Unit
 RAM: Random Access Memory
 ROM: Read-Only Memory
 SAO: Sample Adaptive Offset
 SCC: Screen Content Coding
 SDR: Standard Dynamic Range
 SEI: Supplementary Enhancement Information
 SNR: Signal Noise Ratio
 SPS: Sequence Parameter Set
 SSD: Solid-state Drive
 TU: Transform Unit
 USB: Universal Serial Bus
 VPS: Video Parameter Set
 VUI: Video Usability Information
 VVC: Versatile Video Coding
 WAIP: Wide-Angle Intra Prediction
 What is claimed is:

1. A method of video decoding at a decoder, comprising:
 receiving metadata associated with a coded video bitstream, the metadata including labeling information of one or more objects detected in a first picture that is coded in the coded video bitstream, the labeling information comprising, for an object of the one or more objects, a location of a bounding box surrounding the object in the first picture, and location information that includes a predetermined position value that is outside a picture boundary of a second picture indicating that the bounding box is outside the picture boundary of the second picture that is coded in the coded video bitstream and the object is not shown in the second picture;

decoding the labeling information of the one or more objects in the first picture that is coded in the coded video bitstream; and

applying the labeling information to the one or more objects in the first picture, wherein the object is not shown in the second picture and the bounding box corresponding to the object is not to be displayed in the second picture.

2. The method of claim 1, wherein the metadata is included in a supplementary enhancement information (SEI) message in the coded video bitstream.

3. The method of claim 1, wherein the metadata is included in a file that is separate from the coded video bitstream.

4. The method of claim 1, wherein the labeling information indicates a total number of bounding boxes in the first picture and includes location information and size information of each bounding box, each bounding box being associated with one of the one or more objects in the first picture.

5. The method of claim 1, wherein the labeling information includes object identification information such that a first object appearing in the first picture and another picture coded in the coded video bitstream has the same object identification information in both the first picture and the other picture.

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6. The method of claim 1, wherein the labeling information includes identification information that identifies each of the one or more objects in a video sequence.

7. The method of claim 4, wherein the location information of one of the bounding boxes includes a location offset of the one of the bounding boxes between the first picture and another picture that is coded in the coded video bitstream.

8. The method of claim 1, further comprising:

sending a request to receive the metadata associated with the coded video bitstream.

9. The method of claim 1, wherein the labeling information includes category information that indicates a category for each of the one or more objects.

10. A method of video encoding at an encoder, comprising:

determining labeling information of one or more objects in a first picture that is encoded in a video bitstream, the labeling information comprising, for an object of the one or more objects, a location of a bounding box surrounding the object in the first picture, and location information that includes a predetermined position value that is outside a picture boundary of a second picture indicating that the bounding box is outside the picture boundary of the second picture that is encoded in the video bitstream to indicate that the object is not shown in the second picture; and

encoding metadata associated with the video bitstream, the metadata including the labeling information of the one or more objects in the first picture, wherein the object is not shown in the second picture and the bounding box corresponding to the object is not to be displayed in the second picture.

11. The method of claim 10, wherein the metadata is included in a supplementary enhancement information (SEI) message in the video bitstream.

12. The method of claim 10, wherein the metadata is included in a file that is separate from the video bitstream.

13. The method of claim 10, wherein the labeling information indicates a total number of bounding boxes in the first picture and includes location information and size

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information of each bounding box, each bounding box being associated with one of the one or more objects in the first picture.

14. The method of claim 10, wherein the labeling information includes object identification information such that a first object appearing in the first picture and another picture encoded in the video bitstream has the same object identification information in both the first picture and the other picture.

15. The method of claim 10, wherein the labeling information includes identification information that identifies each of the one or more objects in a video sequence.

16. The method of claim 13, wherein the location information of one of the bounding boxes includes a location offset of the one of the bounding boxes between the first picture and another picture that is encoded in the video bitstream.

17. A method of processing visual media data, the method comprising:

processing a video bitstream of the visual media data according to a format rule, wherein the format rule specifies that:

metadata associated with the video bitstream includes labeling information of one or more objects detected in a first picture that is coded in the video bitstream, the labeling information comprising, for an object of the one or more objects, a location of a bounding box surrounding the object in the first picture, and location information that includes a predetermined position value that is outside a picture boundary of a second picture indicating that the bounding box is outside the picture boundary of the second picture that is coded in the video bitstream and the object is not shown in the second picture;

the labeling information is applied to the one or more objects in the first picture, and

the object is not shown in the second picture and the bounding box corresponding to the object is not to be displayed in the second picture.

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